

SIMULATING EUTOPIA: REVISITING LONG-TERM FAIRNESS WITH OUTCOMES, PERFORMATIVITY, AND DYNAMICS

Anonymous authors

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1. **Why PeRL?** Why the setting follows a PeMDP, or why not? Make it clear. If not, then check if we can at all call it a contribution.
2. **Notations.** Make notations coherent.
3. **Non-intuitive results.**
4. **DM's fairness.** Revisit DM's fairness if it is at all necessary. Rather we can use predictive fairness notion with one-step predictor as discussed earlier.
5. **RL baselines.** Add more baselines. We know DQN does not work, but should try SAC-type actor-critic algos. That means we do not have to justify why only policy gradient for RL baselines.
6. **Discussion on scalability/transferability.** Add a discussion how Eutopia can be adapted to other real-life tasks such as college admission.
7. **Writing.** Make the writing more crisp and direct. There are lots of hanging words, loopy/wrong statements. Add the reactivity discussion in the main text (potentially).
8. **Short-term vs. Long-term.** A discussion on why looking at short-term fairness matters, and why it is in tension with long-term fairness.
9. **Limitations.** Highlight limitations in details.
10. **Ablations.** Try (a) different wealth gap initializations, (b) Different performative strengths, (c) buffer capacity.
11. **Two-sided:** $\text{MaxUtil} - \lambda (\alpha \text{EqualityofOutcome} + (1 - \alpha) \text{EqualityofOpportunity})$ or $\alpha \text{MaxUtil} + (1 - \alpha) \text{EqualityofOutcome/EqualityofOpportunity}$.
12. Reactivity, Lorenz, Reach Rate, non-RL Algorithm, SAC

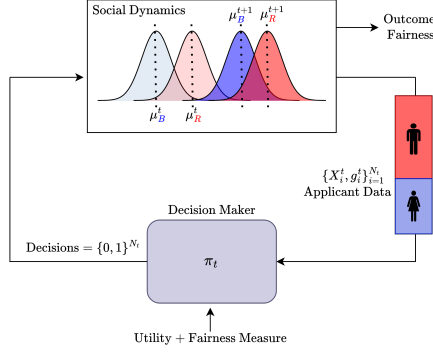


Figure 1: Performative feedback loop of ADM.

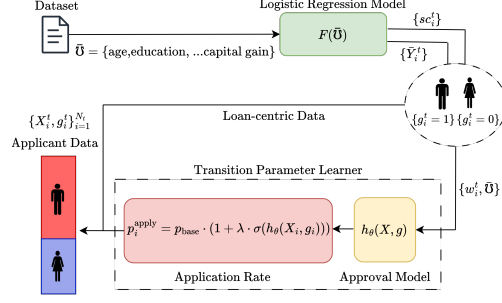


Figure 2: LOANLY: Data pipeline.

ABSTRACT

As AI-driven Decision Makers (ADMs) influence our socioeconomic reality, their roles in both enhancing efficiency and amplifying the social biases have drawn attention. In this paper, we revisit the nuances of long-term ‘fairness’ achievable by an ADM, specifically in the context of a credit lending induced wealth process. The literature on long-term fairness mostly (a) considers passive environments, i.e. the outcome of a predictor does not change the population’s behaviour, and (b) measures bias in terms of disparity in instantaneous predictions rather than the downstream equity. These are not true for modern ADMs, like credit lenders. To address these caveats, we first formalise the wealth dynamics induced by a loan approving ADM interacting with a multi-demographic population as a performative Markov Decision Process with ADM level and social outcome level reward functions. Then, we mitigate the absence of such a performative test-bed by developing **Eutopia**: a lending-process simulator enabled with a novel performative data generator to learn long-term fair strategies. Finally, we test performative and classical RL algorithms with different fairness-aware and utilitarian utilities. Experimental results show that (a) learning with performative dynamics lead to better long-term efficiency and equity, and (b) learning with well-designed fairness-aware utility evaluated on social outcomes induces better efficiency, equity, and inclusivity.

1 INTRODUCTION

Modern AI-driven Decision Makers (ADMs) are not passive. We live in a symbiosis with these digital companions. While they bring efficiency to our high-stake and low-stake daily decisions, they also change our socio-politico-economic behaviours over time. ADMs bring efficiency in applications ranging from credit scoring and loan underwriting (Fuster et al., 2022) to hiring (Raghavan et al., 2020), healthcare triage, and criminal justice (Barocas et al., 2023; Mehrabi et al., 2021; Corbett-Davies et al., 2023). In tandem, they influence how we conduct job or loan applications, how we do medical diagnosis, how we even write papers and conduct research. This reactive effect induced by an algorithm is called *performativity* (Perdomo et al., 2020; Góis et al., 2024; Mandal et al., 2023b) and ADM’s ability to influence human behaviour is measured by its *performative power* (Hardt et al., 2022). This phenomenon invokes:

What are the long-term effects of the performative interactions between AI-driven decision makers and human society, in terms of both efficiency and equity?

Related Work: Algorithmic Fairness in Short and Long-term. Efficiency and equity are not the same. Since ADMs learn from historical data mirroring the existing social inequalities, an unexamined pursuit of predictive accuracy can reproduce, even amplify the structural disadvantages faced by marginalised groups (Angwin et al., 2022; Obermeyer et al., 2019). This has propelled the field of *algorithmic fairness* (Barocas et al., 2023). Researchers developed different metrics to quantify bias, auditors to measure bias induced by ADMs, and algorithmic designs to mitigate bias of predictive systems. The initial works in algorithmic fairness evaluated fairness at a *single decision point*, treating the population of applicants as a fixed, exogenous distribution (D’Amour et al., 2020; Hashimoto et al., 2018). But decision making in real-world is an infinite horizon interactive game.

For example, a loan decision is not a passive observation, rather an intervention. Approving a loan changes the borrower’s wealth that changes their future creditworthiness and also affects dynamics of future applicants. (Liu et al., 2018; Hashimoto et al., 2018; D’Amour et al., 2020) show that learning to maximise *short-term fairness* (e.g. demographic parity in single-step lending) can *reduce* the long-run welfare of disadvantaged groups by denying them the very opportunities (e.g. credit history, asset accumulation). Hence, *long-term fairness is now studied as sequential problem over multiple interactions*.

This has led to Reinforcement Learning (RL) driven study of long-term fairness in ADMs. Jabbari et al. (2017) formalise fairness in *Markov Decision Process* (MDP) setting of RL while requiring that no action make any group worse off relative to a baseline. Subsequent works developed general fairness constraints, and computational-statistical techniques to ensure better efficiency and equity in long-term (Hu & Zhang, 2022; Deng et al., 2022; Ju et al., 2023; Wang et al., 2024). Despite this progress, an overarching assumption persists that the *transition dynamics of the population are exogenous*. The human population is assumed to evolve according to fixed, policy-independent rules, which is not true for modern ADMs. This motivates us to develop a *Performative Reinforcement Learning (PeRL) based formalism for ensuring long-term fairness*.

Contributions and Observations. Our investigation encounters an absence of test-beds to study and demonstrate the impacts of learning algorithms in performative RL. To resolve this, we *first* formalise and develop a data-driven performative simulator, *Eutopia*, for bank lending problem inducing a wealth dynamics among a population of applicants. *Then*, we use this test-bed to study how different (a) *models of learning* (static vs. RL vs. PeRL), (b) *utility designs* (fairness aware vs. utilitarian vs. Rawlsian), and (c) *measures of outcomes* (ADM’s profit vs. social wealth) lead to different trade-offs of long-term efficiency and equity. We address these questions in four layers.

1. Eutopia: A simulator for performative fair lending. We develop a loan simulator grounded in realistic population dynamics that implements (i) a Hawkes-process arrival model— each group’s application rate responds endogenously to observed approval patterns, capturing the behavioural feedback, (ii) a wealth-transition model— approved loans increase borrower wealth conditional on repayment, and (iii) a selective-labels mechanism that censors repayment outcomes for denied applicants (Schnabel et al., 2016; Kallus & Zhou, 2018). The simulator supports both *static* (fixed application rate, MDP) and *performative* modes (endogenous application rate, PeMDP) enabling direct comparison. To our knowledge, *Eutopia* is the first simulator purpose-built for testing performative fairness in long-term.

2. LOANLY: Performative data generator. In order to feed *Eutopia* with realistic application data, we develop LOANLY (Fig. 2). It is a data generation pipeline that constructs synthetic lending datasets from Adult benchmark (Ding et al., 2021)— widely studied to reflect the wealth disparities, group compositions, and temporal dynamics of a credit market. LOANLY provides a standardised on-ramp to test performative RL algorithms in the credit lending setting, without requiring access to proprietary financial data. It includes configurable parameters for group representation ratio, wealth initialisation, default rate, loan gain, interest rate, and performativity strength, and ships with the full validation suite of rule-based heuristics for sanity check. Unlike static dataset releases (Mehrabi et al., 2021), LOANLY encodes performative dynamics enabling study of how a model’s output shape the data to be retrained on.

I. Learning with Performative Dynamics Matters. We compare two families of learned policies: a *static policy gradient* (RL), which trains on trajectories without modelling performative distribution shift, and a *performative policy gradient* (PERL) (Basu et al., 2025). Across all fairness objectives and utilities, PERL achieves lower wealth gaps and inequality ratio than RL in long-term. The gap between the two is particularly significant under outcome-based fairness measure (Table 2). Under Rawlsian utility (RMM), PePG achieves a mean wealth gap 47.2 (vs. 49.1 for PG) under outcome fairness, and the inequality ratio is tightly around $\rho \approx 1$ throughout training. This indicates that performativity is decisive in long-term deployments.

II. Whose Fairness, Which Fairness? A Well-designed Fairness Objective Evaluated on Social Outcomes Matters. We conduct a systematic comparison of long-term fairness objectives: two fairness perspectives (outcome, DM) crossed with four utility functions (utilitarian, social welfare, Rawlsian, Fairness Lagrangian (FL)) (Wu et al., 2025; Jabbari et al., 2017; Hu & Zhang, 2022).

(i) *Outcome-based reward functions achieve the best equity-profit balance.* Under the outcome fairness measure, FL achieves the lowest mean wealth gap, while maintaining a competitive cumulative profit (Table 2). Rawlsian Max-Min (RMM) reward is the most robust to the inequality ratio, keeping it close to 1 across utilities. (ii) *There is no universal winner across metrics.* No single utility function dominates on all the evaluation axes (cumulative profit, wealth gap, social welfare, inequality ratio). This finding suggests that the choice of utility should be treated as a *policy decision* reflecting the lender’s regulatory obligations and social responsibilities. (iii) *Outcome-based fairness training leads to greater inclusivity.* Using group-wise reach rate, i.e. the new applicants being approved per step (Figures 5), we show that outcome-aware Fairness Lagrangian yields a more equitable distribution of loan access than the utilitarian profit objective. For outcome-aware FL, reach rates converge between the two demographic groups within approximately 600 episodes. In brief, the results demonstrate that *outcome-fairness aware utility and performativity-aware algorithm design improve long-term efficiency and equity.*

WRITING FLOW BY SECTIONS

Section 2: Sequential lending as PeMDP. Definition of different quantities of the sequential lending problem → Interaction protocol → Caveat of standard MDP formulation (Why PeMDP?), non-stationarity of p_{apply}^t → definition of reward model, and objective.

Section 3: Eutopia: A Fair Lending Simulator. Why a simulator (D’Amour et al., 2020)? (We can include this in introduction, decide later) → 4 components of Eutopia: 1. Wealth dynamics (what else to write except performative modelling?), 2. Application process (talk about Hawke’s process), 3. LOANLY, 4. Choice of fairness-aware utilities, and fairness metrics (Make them subsections).

Section 4: Performative RL leads to better equity. RL vs PeRL vs heuristic(oracle) performance (discuss advantages of reactivity as a sub-discussion).

Section 5: Outcome based constraints leads to better long-term equity. Equality of Outcome vs Equality of opportunity discussion. Profit is high for Equality of outcome → No bank would be interested to deploy policies that do not fetch *enough* profit. Fairness-utility tradeoff.

Measuring actual outcome of bias matters. To be decided.

2 SEQUENTIAL LENDING AS AN AVERAGE REWARD PERFORMATIVE MDP

In this section, we formalise the sequential credit lending problem (Figure 1) inducing a social wealth dynamics. We consider a population of N applicants, each with a feature vector $\mathbf{x}_i \in \mathbb{R}^d$ and a protected demographic group $g_i \in \{\text{red}, \text{blue}\}$. The learner is equipped with a credit scorer $h_\phi : \mathbb{R}^d \times \{\text{red}, \text{blue}\} \rightarrow \mathbb{R}$ that maps features and group membership to a scalar score $z_i = h_\phi(\mathbf{x}_i, g_i)$, summarising the lender’s estimate of *creditworthiness*. Each applicant requests a loan amount $l_i^t \in \mathbb{R}_{>0}$ and has a binary repayment outcome $Y_i \in \{0, 1\}$, where $Y_i = 1$ denotes successful repayment of the loan. Importantly, Y_i is observed *only for approved applicants*. The lender never learns what would have happened if a denied applicant had have approved a loan due to such *selective label bias*. Additionally, we assume that the first feature coordinate $x_i[1]$ denotes applicant’s wealth.

Interaction Protocol. At each time-step $t \in \mathbb{N}$, a subset of $N_t \leq N$ applicants arrive with their corresponding arrival probability $\{p_{\text{apply}_i^t}\}_{i=1}^{N_t}$. Thus, the **state** is defined over the full context of all arriving applicants as $s_t = \{\mathbf{x}_i, z_i, l_i, g_i\}_{i=1}^{N_t} \in (\mathbb{R}^{d+2} \times \{\text{red}, \text{blue}\})^{N_t}$. Upon observing the state, lender takes a series of binary **actions** $A_t \in \{\perp, \cancel{\perp}\}^{N_t}$, where $\perp$ denotes approval and $\cancel{\perp}$ denial. The expected **transition** of applicant i ’s wealth under a deployed policy π is expressed as:

$$\mathbb{E}[x_i^{t+1}[1]] = (x_i^t[1] + l_i^t) \mathbb{P}_\pi(A_i^t = \perp) \cdot p_{\text{apply}_i^t} + x_i^t[1] \mathbb{P}_\pi(A_i^t = \cancel{\perp}) \cdot p_{\text{apply}_i^t} \quad (1)$$

Why PeMDP? The standard MDP formalism (Sutton et al., 1998) remains oblivious towards the *algorithmic feedback loop* causing *endogenous* shifts in applicants’ behaviour based on the lender’s decisions in this wealth process. Thus, we model long-term fair lending as a *Performative Markov Decision Process* (PeMDP) (Mandal et al., 2023b). Given a policy set $\Pi : \mathcal{S} \rightarrow \Delta_{|\mathcal{A}|}$, the PeMDP is defined as the set of MDPs $\mathcal{M}(\Pi) \triangleq \{\mathcal{M}(\pi) \mid \pi \in \Pi\}$, where each MDP is a tuple $\mathcal{M}(\pi) \triangleq (\mathcal{S}, \mathcal{A}, \mathbf{P}^\pi, r^\pi)$. Note that the transition and reward functions are reactive to the deployed policy $\pi \in \Delta_{|\mathcal{A}|}$ (Rank et al., 2024b; Basu et al., 2025). A *non-performative* setting assumes $p_{\text{apply}_i^t}$ as stationary i.e., applicants applying behaviour remains unaffected by the lender’s historical behaviour. On the contrary, a *performative* setting models the application rate based on the observed approval patterns, specifically as $p_{\text{apply}_i^{t+1}} = f_\lambda(\pi, p_{\text{apply}_i^t}, s_t)$, where f_λ is a parametric response function. Intuitively, a demographic group that is consistently rejected will gradually stop applying, making it harder for the lender to ever learn a fair policy for that group. This layer of feedback, where the policy shapes application behaviour and in turn constrains what future policies can learn, is *absent from prior MDP-based fairness work* (Jabbari et al., 2017; Hu & Zhang, 2022). The bank’s per-step expected reward at interest rate $\tau_{\text{interest}} \in [0, 1]$ is:

$$\mathbb{E}[r_t^\pi] = \sum_{i=1}^{N_t} \left((1 + \tau_{\text{interest}}) \mathbb{P}(Y_i = 1) - \mathbb{P}(Y_i = 0) \right) \cdot \mathbb{P}_\pi(A_i^t = \perp) \cdot l_i^t \cdot p_{\text{apply}_i^t}$$

A policy $\pi : \mathcal{S} \rightarrow \Delta_{|\mathcal{A}|}$ maps the state to lending decisions. As performative distribution shift persists indefinitely, a policy optimised for discounted return may accept long-run harm in exchange for short-run profit. We adopt the average-reward objective: $V^\pi = \sum_{t=0}^T \mathbb{E}_\pi[r_t^\pi]$. Optimising this objective needs reasoning about steady-state behaviour, not just immediate returns.

3 Eutopia: SIMULATING LONG-TERM IMPACT OF ALGORITHMIC DECISIONS

UD: We need opening statements for this section. Maybe we again emphasise on the fact that why a simulator? and also the goal is not to give a universal recipe that ensures long-term fairness, but to re

We propose **Eutopia**, a novel performative simulator realising its four idiosyncratic components: the bank lending PeMDP, a responsive or non-stationary application process modelled using a *Hawkes process* (Hawkes, 1971b), a performative data generator LOANLY feeding it with applicant data, and finally an AI agent trained on different fairness-aware utility functions.

3.1 LOANLY: A PERFORMATIVE DATA GENERATOR

UD: Rewrite this section.

LOANLY is a data-generating pipeline designed for benchmarking algorithms for performative prediction (Perdomo et al., 2020). Unlike static datasets, a performative generator must produce both an initial population *and* the parameters that govern how it responds to decisions over time. LOANLY builds on UCI Adult Income dataset (Becker & Kohavi, 1996) ensuring that the initial population mirrors real demographic and economic heterogeneity. It outputs, for each individual, a wealth $x_i[1]$, a ground-truth creditworthiness label $\bar{Y}_i$, a protected group indicator g_i , and a fixed financial type (l_i, d_i, κ_i) . Here, (i) l_i is the asked amount, (ii) d_i is the default probability, and (iii) κ_i is the gain amount. These together drive the performative feedback once decisions are taken. Following the demographics in Adult, we use gender as the group indicator: the red group corresponds to male applicants and blue group to female applicants, reflecting the empirically documented income and wealth disparities between these groups in the dataset.

Wealth score. A logistic regression is fit to predict $\mathbb{1}[\text{income}_i > \$50\text{K}]$ from age, education, workhours per week, and log-transformed capital gains/losses, yields a continuous score $sc_i = \hat{p}(y_i > \$50\text{K} \mid \bar{U}_i) \in [0, 1]$. Gender is excluded from $\bar{U}$ and enters only as group indicator g_i .

Population construction. Initial wealth is $w_i = 10 + 190(sc_i) \in [\$10\text{K}, \$200\text{K}]$, linearly mapping creditworthiness to a wealth range that reflects the empirical correlation between financial assets and credit risk. The ground-truth label $\bar{Y}_i = \mathbb{1}[sc_i \geq 0.5]$ serves as the fairness reference and is never observed by the agent. Individuals are partitioned by gender into two groups of $N_M = N_F = 3,000$.

Fixed financial types. Each individual is given a loan amount $l_i \sim \mathcal{N}(30, 10^2)$ clipped to $[\$10\text{K}, \$100\text{K}]$ at start. Latent default outcome $d_i \sim \text{Bernoulli}(p_{\text{def}})$, wealth gain on repayment $\kappa_i = l_i(\tau_{\text{inv}} - \tau_{\text{interest}}) = 0.20 l_i$ with $\tau_{\text{inv}} = 0.35$ and $\tau_{\text{interest}} = 0.15$ are fixed throughout. It reflects the partial observability of real lending: latent borrower characteristics are stable but unobserved. The default rate $p_{\text{def}} = (\delta_{\min} + \delta_{\max})/2$ is controlled within the regime $[\delta_{\min}, \delta_{\max}] = [0.02, 0.15]$ for pre-training, and $[0.14, 0.16]$ for evaluation, yielding mean default rates of 8.5% and 15% respectively. The evaluation regime is calibrated to empirically observe subprime lending conditions Demyanyk & Van Hemert (2011), and the resulting distribution shift stress-tests whether policies learned under moderate default risk generalise to a more adverse lending environment.

Approval model and application propensity. $\mathbf{x}$ is the augmented feature set $(w_i, \bar{U})$. LOANLY fits a linear approval model $h_\theta(\mathbf{x}, g) = \theta_g g + \theta_x \mathbf{x} + \theta_{\text{well-behaved}} p_{\text{return upon apply}} + b$, providing the environment with data-driven prior on how wealth and group membership jointly predict creditworthiness. Individual application propensity is modulated as $p_i^{\text{apply}} = p_{\text{base}} \cdot (1 + \lambda \sigma(h_\theta(\mathbf{x}_i, g_i)))$ with $\lambda = 0.5$. We refer to Appendix D.1 for detailed discussion on the data generator.

3.2 WEALTH DYNAMICS

On approval, an individual i gains $\kappa_i = (\tau_{\text{inv}} - \tau_{\text{interest}}) \cdot l_i^t = \tau_{\text{gain}} \cdot l_i^t$ upon repayment or 0 upon default. The positive net return $(\tau_{\text{inv}} > \tau_{\text{interest}})$ makes loans wealth-generating on average. Strategic approval can simultaneously increase bank revenue and group wealth. It creates a non-trivial tension between profit and welfare. Default outcomes are fixed, giving each individual a stable latent credit risk the agent cannot observe directly.

3.3 APPLICATION DYNAMICS: HAWKES ARRIVAL PROCESS

Eutopia is a continual environment with *no state is reset between episodes*: wealth, application arrivals, and cumulative counters all persist, so each episode is a checkpoint on a single continuous trajectory rather than an independent rollout. Otherwise, resetting wealth between episodes would sever the performative feedback loop. Each episode has horizon T with time-step Δt , giving $T \cdot \Delta t$ decision points per episode. Now, we specify the application generation model, and finally action and observation spaces of **Eutopia**.

Hawkes arrival process. As an initial step, we formulate the arrival of applications as a Hawkes process (Hawkes, 1971a) to capture the temporal clustering observed in real credit markets: an approval alleviates the creditworthiness of an individual, which amplifies the creditworthiness of the protected group and encourages further applications, whereas a denied loan or a default causes discouragement. The group-specific intensities are expressed as: $\lambda_g^t = f(\mu_g^t) + \alpha_g \sum_{t_j \in \mathcal{H}_g} e^{-\beta_g(t-t_j)}$, where wealth-driven base $f(\mu_g^t)$ is assumed to be a linear function of the group mean wealth

Fairness Obj.	Equality of Outcome	Equality of Opportunity
SW	$\mu_t^{\text{blue}} + \mu_t^{\text{red}}$	$\text{TPR}_t^{\text{blue}} + \text{TPR}_t^{\text{red}}$
RMM	$\min\{\mu_t^{\text{blue}}, \mu_t^{\text{red}}\}$	$\min\{\text{TPR}_t^{\text{blue}}, \text{TPR}_t^{\text{red}}\}$
FL	$ \mu_t^{\text{blue}} - \mu_t^{\text{red}} $	$ \text{TPR}_t^{\text{blue}} - \text{TPR}_t^{\text{red}} $

Table 1: Fairness constraints across fairness perspectives. Equality of Outcome aims for parity in group wealth. Equality of Opportunity targets the parity in lender’s decisions.

$\mu_g^t = \frac{1}{N_g} \sum_{i=1}^{N_g} x_i^t[1]$. Intuitively, it models the Matthew effect (Merton, 1968) (aka rich getting richer)—inherently wealthy groups tend to and able to apply for more loans, coupling lending decisions with future demands. Parameters are set symmetrically across groups ($\alpha_g = 0.3$, $\beta_g = 2.0$) to isolate policy effects from pre-encoded group asymmetry. The decay $\beta = 2.0$ maintains the excitation in a window of ≈ 3.5 time units.

Application generation and action space. At each timestep, individual i in group g applies with probability $p_{\text{apply}_i}^t = \text{clip}\left(\frac{\lambda_g^t \Delta t}{N_g} \cdot (1 + \lambda \sigma(h_\theta(X_i, g_i)))\right)$, modelling rational self-selection: individuals who perceive a higher approval probability are more likely to apply. The agent outputs a continuous approval probability $a \in [0, 1]$, and the realised decision is Bernoulli(a), enabling gradient-based training without discrete relaxations.

Observation space. The 12-dimensional observation comprises the applicant’s normalised wealth and group indicator, group mean wealth (μ_M, μ_F), Hawkes intensities (λ_M, λ_F), cumulative default rates (ρ_M, ρ_F), the approval model score $\sigma(h_\theta)$, learned coefficients (θ_g, θ_X), and loan amount l_i^t . Bank revenue is $+(1 + \tau_{\text{interest}})l_i^t$ on repayment, $-l_i^t$ on default, and 0 on rejection. For further discussion on the simulator, refer to Appendix D.2. In Appendix E.1, we run seven heuristic policies to validate that Eutopia yields interpretable decisions across fixed rule based policies.

3.4 CHOICES OF FAIRNESS-AWARE UTILITIES

Evaluating policies via simulation with simplified dynamics on their long-term behaviour is a hard task, since building and verifying simulators (even static) that mirror complex real-world scenarios e.g., college admissions (Kleine Buening et al., 2022), is a laborious process. D’Amour et al. (2020) poses the possibility of inverting this problem and use simulation as a way to suggest new fair algorithms using RL as an *open question*. The performance of a learning agent is highly sensitive under different choices of utility definitions. Thus, we ask two questions.

I. Whose Fairness? Profit alone does not tell us whether the lender’s decisions are equitable. We track two group-level statistics for each $g \in \{\text{red}, \text{blue}\}$: reward r_t^g , and the cumulative mean wealth $\mu_t^g = \frac{1}{N_g} \sum_{i=1}^t X_1^{g,i}$. These statistics operationalise two contrasting perspectives on fairness.

Outcome fairness asks whether the long-run wealth distribution is equitable across groups, captured by μ_t^g . *Decision Maker’s (DM) fairness* is to ensure equitable outcome for the lender across groups: whether the bank’s reward derived from red applicants is comparable to that derived from blue applicants. A lender concerned with DM fairness might not help disadvantaged groups as they try to ensure their own business outcomes are not overly concentrated on a single demographic. DM’s self-interested notion of fairness is worth studying as it represents the incentive structure of a profit-driven institution operating under regulatory scrutiny.

We also analyse α -two-sided fairness that interpolates between outcome and DM fairness. The parameter $\alpha \in [0, 1]$ controls the balance: at $\alpha = 0$ the objective reduces to pure DM fairness, while at $\alpha = 1$ it recovers outcome fairness. Refer to Appendix E.3 for details and empirical results.

II. Which Fairness Objective? With the two fairness perspectives, the next task of an agent is to design appropriate objectives to train on. We instantiate all four reward families across the two fairness perspectives in Tables 1, using the following objective functions:

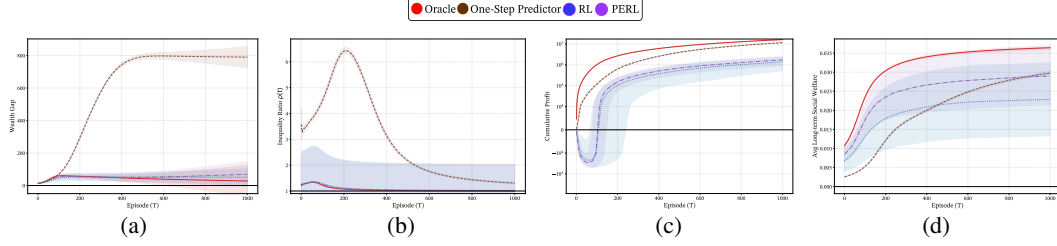


Figure 3: **Learning vs. Heuristics.** Learned agents adapt to the performative feedback loop and maintain equitable wealth trajectories over time, while heuristic policies accumulate profit immediately but produce growing wealth disparities they cannot correct. Refer Figure 9 for more details.

Utilitarian. A utilitarian lender maximises aggregated outcomes with no group-level constraint (Hardt et al., 2016). It encodes a lender’s indifference to equity. Note that outcome fairness has no utilitarian analogue: $\mu_t^{\text{blue}} + \mu_t^{\text{red}}$ oblivious to groups is essentially the social welfare.

Social Welfare. The social welfare lender maximises the sum of group-level outcomes $\mu_t^{\text{blue}} + \mu_t^{\text{red}}$ under outcome fairness (Harsanyi, 1955; Bentham, 1789). This encourages broad participation but does not prevent one group from dominating the objective.

Rawlsian Max-Min. Rawlsian lender prioritises the *disadvantageous* demography (Rawls, 1971) by maximising $\min\{\mu_t^{\text{blue}}, \mu_t^{\text{red}}\}$, and $\min\{r_t^{\text{blue}}, r_t^{\text{red}}\}$ under outcome and DM’s fairness, respectively. This addresses the failure mode of social welfare: no gain for the majority is accepted at the cost of the minority. Under DM’s fairness, the max-min objective ensures the bank’s profit is not lopsidedly generated from one demographic group.

Fairness Lagrangian (FL). Rather than directly optimising a group-level statistic, a Lagrangian regularised utility adaptively penalises group disparities. Under outcome fairness, total wealth is maximised while paying a penalty proportional to the inter-group gap. Under DM fairness, the analogous penalty applies to the bank’s per-group profit. The Lagrangian multiplier $\lambda \geq 0$ encodes a regulatory tolerance for inequality. At $\lambda = 0$, FL collapses to the social welfare or utilitarian objective. As $\lambda \rightarrow \infty$, closing the gap dominates regardless of aggregate welfare (Appendix C). Now, we ask a precise empirical question:

Across the full spectrum from societal to institutional fairness, which objectives produce genuinely equitable long-run outcomes under performative dynamics, and at what cost to profit?

3.5 METRICS OF PROFIT AND EQUITY

Over the multiple episodes, we evaluate policies along two axes: financial performance and distributional equity. We use (i) *macro metrics*: wealth gap, inequality ratio, cumulative profit, and social welfare, which capture population-level outcomes, and (ii) a *micro metric*: reach rate that reveals how intergroup distribution of credit access while being obscured by aggregate statistics.

1. *Wealth Gap* is the difference in mean group wealth, $\Delta_\mu(t) = \mu_M(t) - \mu_F(t)$, where $\mu_g(t)$ denotes the mean wealth of group $g \in \{M, F\}$ at the end of episode t . A value of zero indicates parity, while larger values indicate systematic divergence.
2. *Inequality Ratio* captures relative wealth accumulation rates: $\rho(t) = \frac{\mu_M(t) - \mu_M(0)}{\mu_F(t) - \mu_F(0)}$. $\rho = 1$ indicates proportional growth and $\rho > 1$ indicates the advantaged group is accumulating wealth strictly faster (D’Amour et al., 2020).
3. *Social Welfare* interpreted as the long-term welfare of individual i . It is the asymptotic rate of wealth improvement (Wu et al., 2025): $R_g(t) := \lim_{t \rightarrow \infty} \frac{\mu_g(t) - \mu_g(0)}{t}$.
4. *Cumulative Profit* of the bank is $\text{CP}(t) = \sum_{\tau=1}^t \text{prof}_\tau$, where each approved loan contributes $\tau_{\text{interest}} \cdot l_i$ on repayment and $-l_i$ on default. While contrasted with previous metrics, CP tracks the fundamental trade-off between institutional profit and equity (Hardt et al., 2016).
5. Finally, we measure the micro metric *Reach Rate* for group g as $\text{Reach}_g(t) = |\mathcal{U}_g(t)|/N_g$. It measures the fraction of unique loan recipients in episode t against the group size N_g . Unlike approval rate, reach rate captures whether credit is concentrated among repeating recipients, and in turn, compounds existing advantages.

4 LEARNING HELPS BUT DESIGN CHOICES MATTER

Existing literature (D’Amour et al., 2020) on long-term fairness shows learning agents such as DQN (Mnih et al., 2015) fails to output equitable outcome in the long-run. For our performative simulator Eutopia, we choose policy gradient (Williams, 1992) as the learning agent (RL). For PeRL, we deploy Performative Policy Gradient (PePG) that can achieve near-optimally for an objective while being adaptive to the environment shifts (Basu et al., 2025).

4.1 LEARNING LEADS TO LONG-TERM EQUITY

We first investigate: *whether learning helps to achieve better long-term equity?* Specifically, we compare four algorithms with different models of learning: Oracle: knows the ground truth of defaults, one-step predictor: predicts at every step which applicant would default without considering long-term impacts, RL: learns to optimise long-term utility, and PERL: learns to optimise long-term utility while considering that each decision of DM changes the environment. We run each algorithm for 1000 episodes and report their mean $\pm$ standard deviation of achieved macro metrics in Figure 3.

We observe that learning helps in long-run, and it becomes more effective when we consider a compatible model of learning. While the oracle is the best achievable decision maker with its access to ground truth, we observe that myopic one-step predictor inflates the wealth gap, reduces the social welfare, achieving high profit close to the oracle. In contrast, allowing long-term learning with RL and PERL fast reduces the wealth gap and inequality ratio to the oracle level, while the profit and social welfare reaches closer to the oracle than the myopic one-step predictor. Finally, we observe that though RL and PERL incur comparable profits and wealth gaps, PERL achieves uniformly better social welfare. Thus, the results demonstrate in a multifaceted manner why learning with long-term utility and compatible dynamics model of the world matters for long-term equity. A detailed analysis is reported in Appendix E.2.

4.2 PASSIVE VERSUS PERFORMATIVE MODELS OF RL

We compare same two learning agents across 3 fairness objectives and four reward functions. For each configuration, we report the best-performing reward function per metric in Table 2, and the best fairness objective per reward function in Table 3. Trajectory-level results are in Section E.3.

Observations. 1. First, *outcome fairness paired with FL is the strongest configuration for PERL agent*, yielding the highest profit (174.408 ± 74.473) and social welfare (0.029 ± 0.010) of any configuration in Table 2, alongside an inequality ratio essentially at parity ($\rho = 1.010 \pm 0.049$). 2. *Two-sided fairness is the most effective objective for RL* yielding highest profit (166.211 ± 72.246 under UP) and the closest inequality ratio to parity (0.990 ± 0.051 under RMM) of any RL configuration, while FL achieves the best social welfare (0.030 ± 0.005) with the tightest variance of any welfare result in either table. PERL, however, performs more modestly under two-sided fairness: no reward function shows a clear separation across metrics, suggesting that the performative gradient and passive policy gradient are closely informative when we mix DM-side and outcome-side fairness. 3. *There is no clear winner.* The benefit of performative modelling is fairness-objective-dependent rather than universal. PERL consistently outperforms RL under outcome fairness on profit, welfare, and inequality ratio but the advantage reverses under two-sided fairness where RL dominates on both profit and welfare. For every reward function, PERL and RL select different best fairness objectives on at least one metric in Table 3, confirming that the two agents respond to fairness constraints in qualitatively different ways. The practical implication is that the choice of fairness objective should be made jointly with the choice of learning algorithm, not independently.

4.3 LEARNING FAIR OBJECTIVES LEAD TO BETTER EQUITY

Now, we ask *whether learning with fairness-aware objectives, such as FL or RMM, has any advantage over UP or SW.* We observe that FL or RMM ensure better equity but with some nuances.

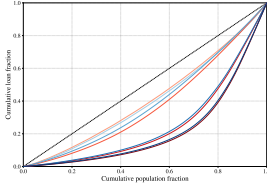
Observations. 1. This contrast is sharpest under outcome fairness, where both agents agree on the ranking. FL consistently achieves the best cumulative profit and social welfare for both RL and PERL (Table 2), while RMM achieves the smallest wealth gap. FL’s Lagrangian penalty creates an incentive to approve broadly across groups, which under performativity, generates more future

Table 2: **Best reward functions per metric across fairness objectives and agent types.** For each fairness objective, we report the fairness objective achieving the best mean value on that metric, together with its mean $\pm$ standard deviation across seeds.

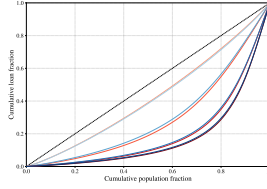
Fairness	Agent	Cumulative Profit ($\times 10^7$)		Wealth Gap		Social Welfare $\bar{R}$		Inequality ρ	
		RF	Mean $\pm$ Std	RF	Mean $\pm$ Std	RF	Mean $\pm$ Std	RF	Mean $\pm$ Std
Outcome	PERL	FL	174.408 $\pm$ 74.473	RMM	47.734 $\pm$ 64.911	FL	0.029 $\pm$ 0.010	FL	1.010 $\pm$ 0.049
	RL	FL	138.420 $\pm$ 86.722	RMM	49.050 $\pm$ 45.353	FL	0.023 $\pm$ 0.013	FL	0.907 $\pm$ 0.261
DM	PERL	UP	112.483 $\pm$ 83.210	UP	54.948 $\pm$ 27.794	FL	0.019 $\pm$ 0.008	UP	0.962 $\pm$ 0.130
	RL	UP	116.322 $\pm$ 107.624	RMM	57.980 $\pm$ 25.737	UP	0.024 $\pm$ 0.014	RMM	0.936 $\pm$ 0.259
Two-Sided	PERL	RMM	123.561 $\pm$ 90.776	RMM	34.017 $\pm$ 71.174	SW	0.022 $\pm$ 0.014	UP	0.874 $\pm$ 0.291
	RL	UP	166.211 $\pm$ 72.246	UP	29.559 $\pm$ 66.819	FL	0.030 $\pm$ 0.005	RMM	0.990 $\pm$ 0.051

Table 3: **Best fairness objective per metric across reward functions and agent types.** For each reward function, we report the fairness objective achieving the best mean value on that metric, together with its mean $\pm$ standard deviation across seeds.

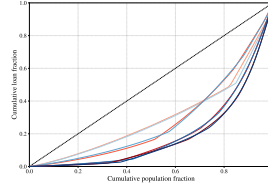
Reward	Agent	Cumulative Profit ($\times 10^7$)		Wealth Gap		Social Welfare $\bar{R}$		Inequality ρ	
		Fairness	Mean $\pm$ Std	Fairness	Mean $\pm$ Std	Fairness	Mean $\pm$ Std	Fairness	Mean $\pm$ Std
UP	RL	Two-S.	166.211 $\pm$ 72.246	Two-S.	29.559 $\pm$ 66.819	Two-S.	0.028 $\pm$ 0.007	Two-S.	0.996 $\pm$ 0.054
	PERL	DM	112.483 $\pm$ 83.210	Two-S.	36.084 $\pm$ 26.702	Two-S.	0.017 $\pm$ 0.015	Two-S.	0.874 $\pm$ 0.291
SW	RL	Two-S.	164.898 $\pm$ 88.075	Two-S.	51.991 $\pm$ 72.339	Two-S.	0.027 $\pm$ 0.010	Outcome	0.978 $\pm$ 0.061
	PERL	Outcome	140.769 $\pm$ 101.080	Two-S.	35.610 $\pm$ 70.530	Outcome	0.024 $\pm$ 0.013	Outcome	1.056 $\pm$ 0.191
RMM	RL	Outcome	93.306 $\pm$ 82.358	Two-S.	48.777 $\pm$ 31.473	Two-S.	0.017 $\pm$ 0.003	DM	0.936 $\pm$ 0.259
	PERL	Two-S.	123.561 $\pm$ 90.776	Two-S.	34.017 $\pm$ 71.174	Two-S.	0.020 $\pm$ 0.014	DM	0.990 $\pm$ 0.064
FL	RL	Two-S.	147.142 $\pm$ 83.210	Two-S.	35.822 $\pm$ 78.928	Two-S.	0.030 $\pm$ 0.005	Outcome	0.907 $\pm$ 0.261
	PERL	Outcome	174.408 $\pm$ 74.473	Two-S.	50.442 $\pm$ 33.648	Outcome	0.029 $\pm$ 0.010	DM	1.016 $\pm$ 0.038



(a) RMM



(b) FL



(c) UP

Figure 4: **Lorenz curves under DM fairness (PERL).** Red: male group, Blue: female group. Curves shown at episodes 100, 400, 800, and 1000 (light to dark). Dashed line is perfect equality.

applicants and thus more long-run profit. RMM concentrates its fairness effort on lifting the weaker group’s absolute wealth level rather than maximising aggregate returns. 2. Under DM’s fairness, where SW has no analogue and UP dominates, both agents produce the largest wealth gaps among all configuration. This reinforces that objectives blind to outcome dynamics fails to yield equitable outcome in long-run. 3. The picture under two-sided fairness complicates any universal ranking. UP achieves the highest profit for RL while RMM achieves the best wealth gap, and the spread across reward functions is the widest for any constraint (Table 3).

To conclude, objectives that explicitly target inter-group equity (FL, RMM) produce better long-run equity outcomes without systematically sacrificing profit.

5 MEASURING ACTUAL OUTCOME OF BIAS MATTERS

Metrics such as wealth gap and inequality ratio measure fairness between groups, but they are oblivious towards how credit access is distributed within groups. A policy can achieve near-perfect group-level parity while systematically concentrating loans among the same high-wealth individuals. To surface this, we examine two complementary views: *Lorenz curves* (Figure 4, 17) to track cumulative loan concentration within groups over deployment, and *reach rate* plots (Figure 5) for tracking the fraction of unique recipients per episode relative to the full group size. Cumulative loan distribution histograms, which reveal the within-group skewness of total loans received in short-term versus the full deployment horizon, are reported in Appendix E.3.4.

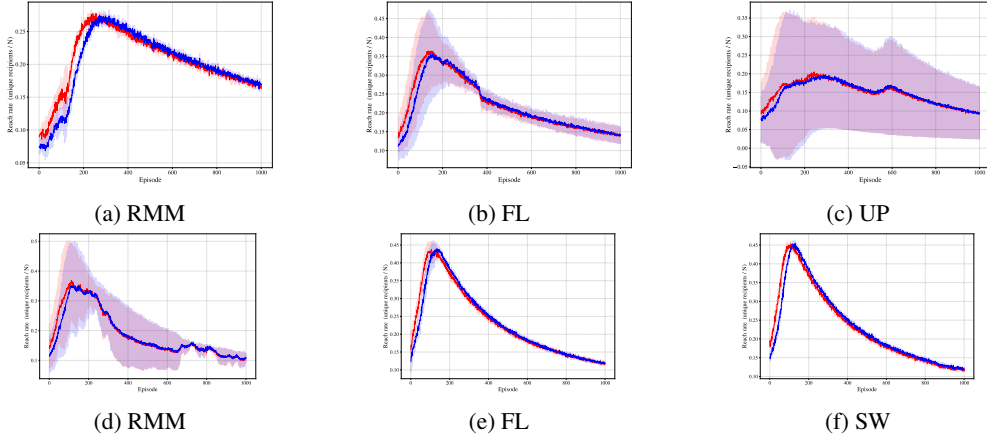


Figure 5: **Reach rate under DM’s (a-c) and outcome fairness (d-f) (PERL).** Red: male group, blue: female group. Shaded regions show standard deviation across seeds.

Observations. 1. *Matthew Effect is Universal.* Across all configuration, Lorenz curves move away from uniform as deployment progresses and reach rates peak then decline. Early in deployment, wealth is relatively uniformly distributed and the policy approves a broad cross-section of applicants. As approved individuals accumulate wealth, they become more likely to apply and be approved again in future, while others fall progressively further behind. 2. *DM’s Fairness* yields the largest wealth gaps regardless of agent type or reward function, with both RL and PERL agents. This is a structural consequence of constraining per-step profit disparity between groups places without any penalty on wealth divergence. Under FL, the male and female loan distributions are clearly separated throughout deployment, confirming that penalising profit disparity does not prevent the female group from being systematically excluded from credit access as wealth stratifies. Under RMM, the two groups appear more closely interleaved, but this reflects a shared reduction in total loan volume rather than genuine redistribution toward lower-wealth applicants. 3. *Outcome Fairness.* Outcome fairness achieves higher reach rates than DM’s fairness across all reward functions, but at the cost of sharper within-group concentration. FL and SW show near-perfect group parity, yet their cumulative loan histograms are the most concentrated, consistent with their shared objective of maximising total group wealth rewarding repeated lending to the already-wealthy. A key tension emerges under RMM: switching from DM to outcome fairness reduces the between-group wealth gap at the aggregate level, but worsens within-group equity for the disadvantaged group.

6 DISCUSSION, LIMITATIONS AND OPEN QUESTIONS

We revisit the long-term fairness achieved by an algorithmic decision maker interacting with a population. We ask specific questions: (a) whether learning matters to ensure long-term equity, (b) whether considering performative effect of decisions lead to better equity, (c) how different fairness objectives influence different metrics of profit and equity achieved by the DM and the population, (d) whether considering the true impact of decisions in terms of social outcomes lead to better long-term equity both in terms of aggregated measures over different demographics and inside the demography itself. In this direction, we develop a novel performative simulator and data generator, and numerically address these questions and their inherent nuances. This shows learning with a proper model of population dynamics and well-designed fairness objectives evaluated on social outcomes matter for ensuring long-term equity. While our goal is not to mimic the real world, the performative feedback, wealth dynamics, and behavioural responses modelled in [Eutopia](#) already produce outcomes that are non-trivial and difficult to anticipate without simulation (D’Amour et al., 2020). This motivates development of new algorithmic frameworks to ensure long-term equity while acknowledging the multifaceted nature of this problem and tensions between different measures and perspectives of fairness.

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If you’d like to, you may include a section for author contributions as is done in many journals. This is optional and at the discretion of the authors.

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Appendix

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A DETAILED RELATED WORKS

The study of fairness in machine learning has evolved from static, snapshot-based metrics toward a dynamic understanding of how algorithms reshape the populations they serve.

Static Fairness and Its Limitations. Early foundational work established formal definitions of algorithmic fairness, including individual fairness (Dwork et al., 2012) and group fairness criteria such as equalized odds and calibration (Hardt et al., 2016). A key theoretical insight from this era is that many natural fairness criteria are mutually exclusive except in degenerate cases (Chouldechova, 2017; Kleinberg et al., 2016). More critically, these definitions treat fairness as a property of a single decision point, ignoring feedback dynamics inherent in repeated deployment. Short-term fairness metrics in lending can reduce the long-term welfare of disadvantaged groups by denying them opportunities to build credit history (Liu et al., 2018). Similarly, static fairness constraints create feedback loops: a policy trained on historically biased data may perpetuate exclusion, as the population distribution at time $t+1$ depends on decisions made at time t (Wyllie et al., 2024). These findings motivate reasoning about fairness across extended temporal horizons.

Sequential Fairness in MDPs. To address temporal dependencies, researchers recast fair decision-making as a sequential problem. Initial work in the bandit setting (Joseph et al., 2016; Nanda et al., 2021) introduced fairness constraints that account for shifting outcome distributions. The natural generalization is to Markov Decision Processes (MDPs), where the agent must balance immediate fairness requirements against long-term consequences encoded in state dynamics. Subsequent work developed tractable algorithms: Lyapunov-based methods for high-probability constraint satisfaction (Hu & Zhang, 2022), primal-dual approaches for online fair policy learning (Deng et al., 2022; Wang et al., 2024), and sample-efficient methods for unknown dynamics (Ju et al., 2023). A common assumption across this literature, however, is that transition dynamics are exogenous—the environment evolves according to fixed, policy-independent rules. This assumption breaks down when the population adapts its behavior in response to the deployed policy.

Performative Fairness and Long-term Welfare. The intersection of long-term fairness and performative prediction (Perdomo et al., 2020) represents the current frontier. Performative prediction recognizes that deploying a model changes the distribution it was trained on: a loan approval model affects who applies for loans, and a recidivism predictor influences defendant behavior. Furthermore, a foundational analysis of different policy objectives demonstrates the steady-state distribution of welfare across demographic groups (Wu et al., 2025). It is important to note that the choice of objective has profound implications for long-run inequality. However, pre-existing frameworks do not account for the *performative* shift in transition dynamics that arises when populations respond strategically to deployed policies. Recent theoretical advances have begun to address this gap by establishing conditions for performative stability in discrete MDPs (Mandal et al., 2023a; Rank et al., 2024a), as well as by developing policy gradient methods with convergence guarantees under performative distribution shift (Basu et al., 2025).

B REACTIVITY OF A LEARNER

We propose a structural classification of reactivity for a learning algorithm operating in a performative environment. Standard analyses of long-term fairness compare policies by their cumulative profit, wealth gap, social welfare, or inequality ratio, but none of these isolate how a learner lets information from the environment propagate into its decision rule. Reactivity is the missing axis: it asks whether the learner’s responsiveness coefficient is itself updated as the environment shifts, or remains pinned to its training-time value.

Model. We model the learner-environment interaction as a coupled linear system with exogenous input Z_t and i.i.d. Gaussian noises $\xi_t \sim \mathcal{N}(\mu_\xi, \sigma_\xi^2)$, $\eta_t \sim \mathcal{N}(\mu_\eta, \sigma_\eta^2)$:

$$X_{t+1} = \alpha X_t + \beta Y_t + \delta Z_t + \xi_t, \quad (2)$$

$$Y_{t+1} = a Y_t + b X_t + \eta_t, \quad (3)$$

with $|\alpha|, |a| < 1$ for boundedness. X_t denotes the environment state and Y_t the learner’s decision variable.

Trajectory unrolling. Substituting equation 2 into equation 3 recursively and gathering terms by source yields a three-part decomposition,

$$Y_{t+1} = a Y_t + \underbrace{\beta \sum_{s=0}^{t-1} \alpha^{t-1-s} b_s Y_s}_{\text{Past}} + \underbrace{\sum_{s=0}^{t-1} \alpha^{t-1-s} b_s \delta_s Z_s + b_0 \alpha^t X_0}_{\text{Data}} + \underbrace{\sum_{s=0}^{t-1} \alpha^{t-1-s} b_s \xi_s + \eta_t}_{\text{Noise}}. \quad (4)$$

The Past term sums contributions from earlier learner states; the Data term is the running sum of inputs $\{Z_s\}_{s < t}$ weighted by the per-step responsiveness b_s , plus the propagated initial condition $b_0 \alpha^t X_0$; the Noise term collects the noise from both the environment and the learner. Reactivity is governed by the Data term, and specifically by where b_s enters the temporal sum.

Local sensitivities. The two channels through which information moves are the partial derivatives $\delta_t \triangleq \frac{\partial X_t}{\partial Z_t}$, $b_t \triangleq \frac{\partial Y_t}{\partial X_t}$, both allowed to be time-varying in equation 2–equation 3. The qualitative behaviour of the learner is determined by whether b_t remains pinned to its initial value b_0 or evolves with the dynamics.

Definition 1 (Reactivity classes). *Under equation 2–equation 3, a learner is (i) **non-reactive** if $b_t \equiv 0$; (ii) **passive** if $b_t = h(X_t)$ for some t -independent function h ; (iii) **reactive** if $b_t = h_t(X_t)$ with h_t depending genuinely on t .*

Empirical evolution. For each agent we set $b_t = \nabla_{\mu} \pi_t$, the gradient of the policy’s expected approval probability with respect to the two group-mean wealths ($\mu^{\text{red}}, \mu^{\text{blue}}$), estimated by autograd on the per-episode batch of states seen by the agent. We run each (agent, reward, fairness) combination for $T = 1000$ episodes across 5 seeds. Figures 6 and 7 show the per-episode magnitude $\|b_t\|$ and its running sum $\sum_{s \leq t} \|b_s\|$, one panel per reward family and one curve per fairness objective, for the combinations of Section 3.4 (UP $\times$ DM only; SW $\times$ outcome only; RMM, FL $\times$ {outcome, DM}). Three observations follow.

First, PERL ends with a higher running sum than RL in the combinations where the policy is most responsive overall (Figure 7). Under UP $\times$ DM, PERL reaches $\approx 1.5 \times 10^{-3}$ at $t = 1000$ versus $\approx 0.2 \times 10^{-3}$ for RL; under RMM $\times$ outcome, PERL ends roughly twice as high. This matches the *reactive* class of Definition 1: PERL keeps updating its policy during deployment, while RL’s frozen weights leave the response function fixed. The exception is FL $\times$ DM, where RL ends higher than PERL.

Second, the per-episode curve decays toward zero in every configuration (Figure 6), with most sensitivity acquired in the first 100–200 episodes; past this point, $\|b_t\|$ stays close to zero for the rest of deployment.

Third, the size of the PERL/RL gap depends on the reward and fairness pair: largest under UP $\times$ DM and RMM $\times$ outcome, small under SW $\times$ outcome and RMM $\times$ DM, and flipped under FL $\times$ DM.

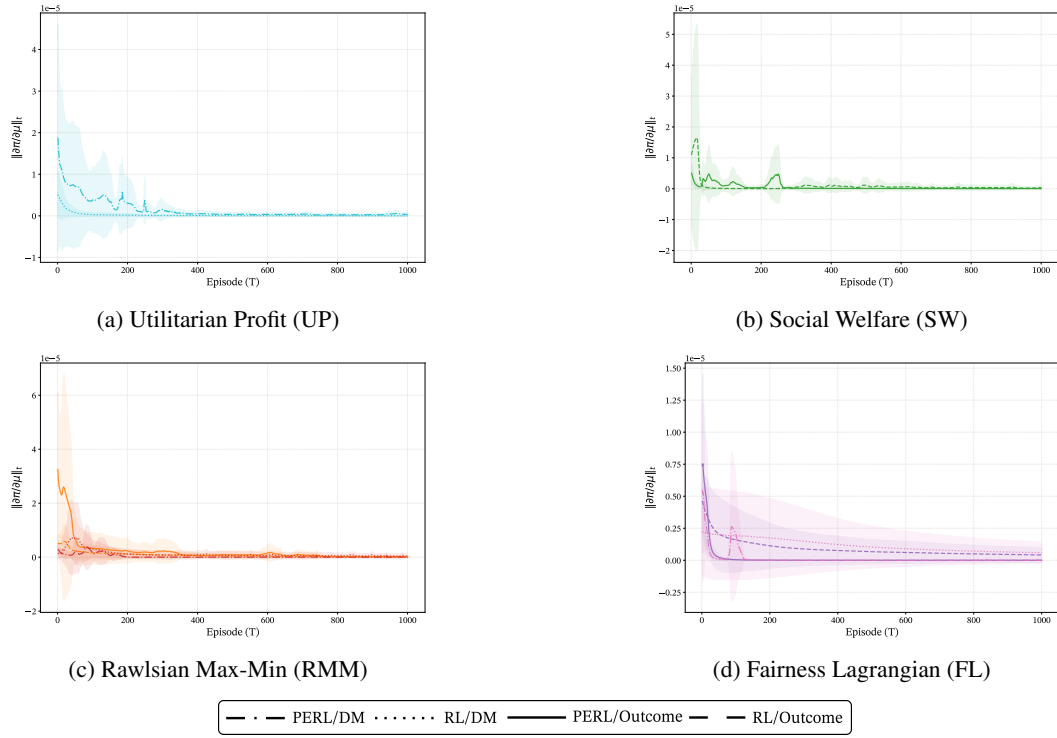


Figure 6: **Episodic reactivity:** $\|b_t\|$ over $T = 1000$ deploy episodes, one panel per reward family. Each panel overlays the available fairness objectives. Mean $\pm$ std across seeds.

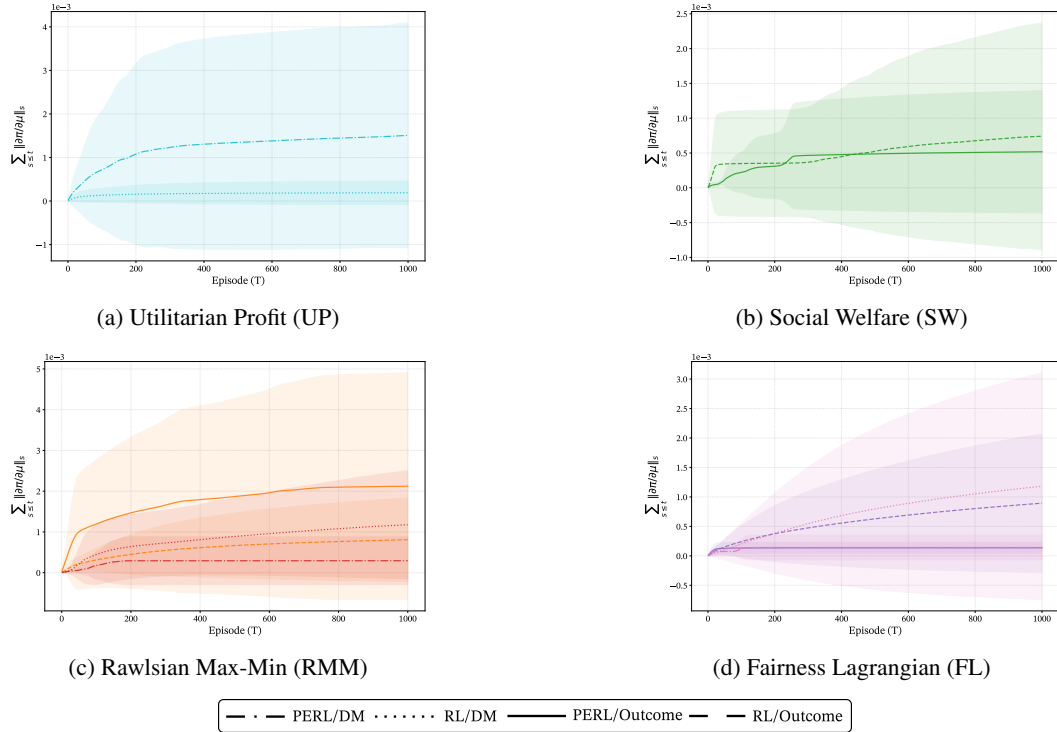


Figure 7: **Cumulative reactivity:** Running sum $\sum_{s \leq t} \|b_s\|$ of reactivity over $T = 1000$ deploy episodes, one panel per reward family. Each panel overlays the available fairness objectives. Mean $\pm$ std across seeds.

C PSEUDO-CODES

Algorithm 1 Loan Simulator Interaction Protocol

```

1: Notation:
2:  $\mathcal{R}, \mathcal{B}$ : Red (male) and Blue (female) populations of size  $N_R, N_B$ 
3:  $X_i^g$ : wealth of individual  $i$  in group  $g \in \{\mathcal{R}, \mathcal{B}\}$ 
4:  $\mu_t^g = \frac{1}{N_g} \sum_i X_i^g$ : mean group wealth at time  $t$ 
5:  $\rho^g = \text{total\_defaults}^g / \max(\text{total\_loans}^g, 1)$ : running default rate
6:  $\kappa_i$ : individual wealth gain on successful repayment
7:  $\delta_i \in \{0, 1\}$ : default indicator,  $\delta_i \sim \text{Bernoulli}(p_i^{\text{def}})$ 
8:
9: Initialise  $X_i^g \leftarrow \text{data}$ ,  $\mu^g \leftarrow \mathbb{E}[X^g]$ ,  $\lambda_0^g \leftarrow f(\mu^g)$ 
10: for episode  $e = 1, \dots, E$  do
11:   // Soft-reset (performative): carry over wealth, reset episode counters
12:    $\mu^g \leftarrow$  carry-over from previous episode
13:   Reset  $\text{episode\_loans}^g$ ,  $\text{episode\_applications}^g \leftarrow 0$ 
14:   for timestep  $t = 0, \Delta t, 2\Delta t, \dots, T$  do
15:     // — Hawkes arrival process —
16:      $\lambda^g(t) \leftarrow \underbrace{\text{toward } f(\mu^g)}_{\text{base rate}} + \alpha^g \sum_{t_k < t} e^{-\beta^g(t-t_k)}$ 
17:     where  $f(\mu) = \max(0.5, 2.0 + 0.05\mu)$ 
18:     for each group  $g$  do
19:        $p_i^{\text{app}} \leftarrow \frac{\lambda^g(t) \Delta t}{N_g} \cdot (1 + s^g \cdot p_i^\theta)$  for each  $i \in g$ 
20:       Draw applicants  $\mathcal{A}_t^g \leftarrow \{i : u_i < p_i^{\text{app}}, u_i \sim \mathcal{U}[0, 1]\}$ ,  $|\mathcal{A}_t^g| \leq 10$ 
21:     end for
22:     // — Policy decision —
23:     Agent observes  $o_t = (\lambda^R(t), \lambda^B(t), \mu^R, \mu^B, \rho^R, \rho^B, X_i, S_i, p_i^\theta, L_i)$ 
24:     Agent outputs  $\pi \leftarrow \pi_\theta(o_t) \in [0, 1]$  (approval probability)
25:     for each applicant  $i \in \mathcal{A}_t^g$  do
26:        $a_i \sim \text{Bernoulli}(\pi)$ 
27:       if  $a_i = 1$  (approved) then
28:          $\delta_i \sim \text{Bernoulli}(p_i^{\text{def}})$ 
29:          $X_i^g \leftarrow X_i^g + \kappa_i \cdot \mathbf{1}[\delta_i = 0]$ 
30:         Append  $t$  to Hawkes history  $\mathcal{H}^g$ 
31:         Update  $\text{total\_loans}^g$ ,  $\text{total\_defaults}^g$ 
32:       end if
33:     Update  $\text{total\_applications}^g$ 
34:   end for
35:    $\mu^g \leftarrow \frac{1}{N_g} \sum_i X_i^g$  (performative state update)
36:   Compute reward  $r_t$  (see Algorithms 2–4)
37: end for
38: Agent updates  $\theta$  via policy gradient on  $\{r_t\}_{t=0}^T$ 
39: end for

```

Algorithm 1 specifies the shared interaction protocol. Each episode carries wealth forward from the previous one (a performative soft-reset), while loan and application counters reset. Applicants arrive via a Hawkes process whose base rate depends on group-mean wealth μ^g ; the agent observes group-level statistics and applicant features and outputs an approval probability $\pi_\theta(o_t)$. Approved applicants default stochastically, wealth updates on successful repayment, and the resulting group means feed back into the next timestep’s arrival rate and observation — the performative loop underlying the simulator. Rewards are computed per step and accumulated for a policy-gradient update at episode’s end.

Algorithms 2–4 instantiate the reward under the three fairness regimes of Section 3.4 — DM (lender-side profit only), Social/Outcome (group wealth only), and Two-Sided (a blend of both via α) — each crossed with four aggregation rules: Utilitarian and Social Welfare (sum across groups), Rawl-

Algorithm 2 Fairness objective — Decision Maker (DM) Fairness

1: **Per-step bank profit** (for current applicant i with loan L_i , interest rate ι):

$$r_t^\pi = \sum_{i=1}^{N_t} \pi_i \cdot \left[(1 - p_i^{\text{def}}) \cdot L_i \cdot \iota - p_i^{\text{def}} \cdot L_i \right] = \sum_{i=1}^{N_t} \pi_i (1 - (1 + \iota) L_i p_i^{\text{def}})$$

2: **Per-group profit rate** (running estimate):

$$r^g = \bar{L}^g \cdot \left[(1 - \rho^g) \cdot \iota - 2\rho^g \right], \quad \rho^g = \frac{\text{total_defaults}^g}{\max(\text{total_loans}^g, 1)}$$

3:

4: **Utilitarian Profit:** $r_t \leftarrow r_t^\pi$

5:

6: **Social Welfare:** $r_t \leftarrow 0$ (undefined under DM constraint)

7:

8: **Rawlsian Max-Min:** $r_t \leftarrow \min(r^R, r^B)$

9:

10: **Fairness Lagrangian:**

$$r_t \leftarrow r_t^\pi - \lambda \cdot |r^R - r^B|$$

Algorithm 3 Fairness objective — Social / Outcome Fairness

1: **Core signal** — mean group wealth (performative state):

$$\mu_t^g = \frac{1}{N_g} \sum_{i \in g} X_i^g(t)$$

2: updated after each approval via $X_i^g \leftarrow X_i^g + \kappa_i \cdot \mathbf{1}[\delta_i = 0]$

3:

4: **Running approval rates** (cumulative):

$$\text{app}^g = \frac{\text{total_loans}^g}{\max(\text{total_applications}^g, 1)}$$

5:

6: **Utilitarian Profit:**

7: $r_t \leftarrow \mu_t^R + \mu_t^B$

8: *Maximise total societal wealth; no per-step profit signal, no fairness penalty.*

9:

10: **Social Welfare:**

11: $r_t \leftarrow \mu_t^R + \mu_t^B$

12: *Identical to Utilitarian under the social constraint.*

13:

14: **Rawlsian Max-Min:**

15: $r_t \leftarrow \min(\mu_t^R, \mu_t^B)$

16: *Maximise the wealth of the worse-off group; agent must lift the lagging group.*

17:

18: **Fairness Lagrangian:**

19: $r_t \leftarrow \mu_t^R + \mu_t^B - \lambda \cdot |\mu_t^R - \mu_t^B|$

20: *Penalise wealth gap; λ trades off total wealth vs. equality.*

Algorithm 4 Fairness objective — Two-Sided Fairness

1: **Per-step bank profit** (for current applicant i with loan L_i , interest rate ι):

$$r_t^\pi = \pi_i \cdot \left[(1 - p_i^{\text{def}}) \cdot L_i \cdot \iota - p_i^{\text{def}} \cdot L_i \right] \quad (5)$$

2: **Blending parameter** $\alpha \in [0, 1]$ trades off profit against group welfare.

3:

4: **Utilitarian Profit:**

$$r_t \leftarrow (1 - \alpha) \cdot r_t^\pi + \alpha \cdot \frac{\mu_M(t) + \mu_F(t)}{2\bar{L}} \quad (6)$$

5:

6: **Social Welfare:**

$$r_t \leftarrow \frac{r_t^\pi + \mu_M(t) + \mu_F(t)}{1 + N} \quad (7)$$

7:

8: **Rawlsian Max-Min:**

$$r_t \leftarrow (1 - \alpha) \cdot r_t^\pi + \alpha \cdot \min(\mu_M(t), \mu_F(t)) \quad (8)$$

9:

10: **Fairness Lagrangian:**

$$r_t \leftarrow (1 - \alpha) \cdot r_t^\pi - \alpha \cdot |\mu_M(t) - \mu_F(t)| \quad (9)$$

sian Max-Min (minimum), and Fairness Lagrangian (penalized gap). This factorization crosses every fairness constraint with every aggregation rule while holding the simulator dynamics fixed.

D EXPERIMENTAL SETTINGS

D.1 LOANLY: DATA PIPELINE DETAILS

LOANLY is a data-generating pipeline designed specifically for benchmarking algorithms in the performative prediction setting (Perdomo et al., 2020). Unlike static datasets, a performative data generator must produce not only an initial population but also the parameters that govern how that population responds to decisions over time, which includes how wealth evolves, application behaviour shifts, and the distribution the learner faces at episode $t + 1$ depends on the decisions made at episode t . LOANLY grounds this feedback structure in the UCI Adult Income dataset (Becker & Kohavi, 1996), ensuring that the initial population reflects real demographic and economic heterogeneity rather than synthetic assumptions. Concretely, it produces for each individual a wealth w_i , a ground-truth creditworthiness label $\bar{Y}_i$, a group indicator g_i , and a fixed financial type (l_i, d_i, κ_i) encoding their loan amount, default rate, and wealth gain, the parameters that drive the performative feedback once decisions begin. We list all choices and parameters below.

Wealth Score $F(\bar{U})$. We derive a continuous score $sc_i \in [0, 1]$ by fitting a logistic regression to predict $\mathbf{1}[\text{income}_i > \$50\text{K}]$ from five features: age, education, hours worked per week, and log-transformed capital gains and losses. The output $sc_i = \hat{p}(y_i > \$50\text{K} \mid \bar{U}_i)$ provides a smooth, interpretable proxy for credit risk. Gender is not included in $\bar{U}$; it enters only as the group indicator g_i .

Population construction. Each individual’s initial wealth is set as $w_i = 10 + 190 \cdot sc_i \in [10, 200]$, linearly mapping creditworthiness onto a wealth range that reflects the empirical correlation between financial assets and credit risk. The ground-truth approval label $g_i = \mathbf{1}[sc_i \geq \tau]$ (default $\tau = 0.5$) serves as the fairness reference and is not visible to the agent. Individuals are partitioned by sex into two groups ($g_i = 1$ male, $g_i = 0$ female); the number sampled per group, N_M and N_F , is a user-specified input (default $N_M = N_F = 3,000$).

Transition Parameter Learner. The learner fits an approval model $h_\theta(\mathbf{x}, g) = \theta_g g + \theta_X \mathbf{x} + b$ on the processed population, giving the environment a data-driven prior on how wealth and group membership jointly predict creditworthiness. Each individual is also assigned a fixed financial type (l_i, d_i, κ_i) drawn once at initialisation and held constant across all episodes, which reflected the partial observability of real lending, where latent borrower characteristics are stable but unobserved. Loan amounts $l_i \sim \mathcal{N}(30, 10^2)$ are clipped to $[10, 100]$; default outcomes $d_i \sim \text{Bernoulli}(p_{\text{def}})$; and wealth gains on repayment $\kappa_i = l_i(\tau_{\text{inv}} - \tau_{\text{interest}})$ with $\tau_{\text{inv}} = 0.35$, $\tau_{\text{interest}} = 0.15$.

The default rate $p_{\text{def}} = (\delta_{\min} + \delta_{\max})/2$ is controlled by a configurable regime: $[\delta_{\min}, \delta_{\max}] = [0.02, 0.15]$ for pre-training (broad, lower-risk) and $[0.14, 0.16]$ for evaluation (narrow, higher-risk), inducing a controlled distribution shift that stress-tests whether learned policies generalise. All results in the main text use the evaluation regime. Application propensity is modulated per-individual as $p_i^{\text{apply}} = p_{\text{base}} \cdot (1 + \lambda \cdot \sigma(h_\theta(\mathbf{x}_i, g_i)))$ with $\lambda = 0.5$, coupling application rates to perceived approval probability.

D.2 Eutopia: DETAILS

Eutopia is the performative loan-simulator built on top of the LOANLY data generator. Its defining property is that *no state is reset between episodes* i.e. wealth, Hawkes event history, and cumulative counters all persist, so each episode is a checkpoint on a single continuous trajectory rather than an independent rollout. This is a deliberate design choice: resetting wealth between episodes would break the performative feedback loop, as the distribution the agent faces at episode $t + 1$ is precisely the one shaped by its decisions at episode t .

Population. $N_M = N_F = 3,000$ applicants per group. This size is large enough to produce stable group-level statistics (mean wealth, approval rates) while remaining computationally tractable for multi-seed runs. Initial wealth $w_i = 10 + 190 sc_i \in [10, 200]$ and ground truth $\bar{Y}_i = \mathbf{1}[w_i \geq 0.5]$ are produced by LOANLY (Section 3.1).

Time. Episode horizon $T = 1000$, timestep $\Delta t = 0.5$, giving 2000 decision timesteps per episode. This provides enough loan decisions per episode for meaningful policy gradient estimates while keeping individual episodes short enough to checkpoint frequently. Continuous time t advances monotonically across episodes, so the Hawkes history and wealth state accumulate over the full training run.

Hawkes process. Application arrivals follow a Hawkes process rather than a homogeneous Poisson process to reflect the temporal clustering of loan applications observed in real credit markets i.e. an approval signals creditworthiness and encourages others to apply. Group-specific intensities are $\lambda_g(t) = f(\mu_g) + \alpha_g \sum_{t_j \in \mathcal{H}_g} e^{-\beta_g(t-t_j)}$, where $f(\mu) = \max(0.5, 2 + 0.05\mu)$ ties the base application rate to current group mean wealth — wealthier groups apply more frequently, creating a coupling between the policy’s lending decisions and future demand. Parameters $\alpha_R = \alpha_B = 0.3$ and $\beta_R = \beta_B = 2.0$ are set symmetrically across groups to isolate the effect of the policy rather than pre-encoding group-level asymmetry. The decay $\beta = 2.0$ confines excitation to a local time window of ≈ 3.5 units, preventing unbounded intensity growth. Events older than $-\log(10^{-6})/\beta \approx 6.9$ time units are pruned for memory management, as their contribution is negligible.

Wealth transition. On approval, individual i ’s wealth increases by $\kappa_i = l_i(\tau_{\text{inv}} - \tau_{\text{interest}}) = 0.20 l_i$ if they repay, or 0 if they default, where $r_{\text{inv}} = 0.35$ and $\tau_{\text{interest}} = 0.15$. The positive net return $r_{\text{inv}} > r_{\text{int}}$ ensures that loans are wealth-generating for borrowers on average, making the tension between profit and welfare non-trivial - an agent that approves strategically can simultaneously increase bank revenue and group wealth. Wealth is measured in \$K. Default outcomes $d_i \sim \text{Bernoulli}(p_{\text{def}})$ are pre-drawn per individual and fixed across episodes, so each individual has a stable latent credit risk that the agent cannot directly observe.

Application generation. At each timestep, individual i in group g applies independently with probability $p_i^{\text{apply}} = \text{clip}\left(\frac{\lambda_g(t) \Delta t}{N_g} \cdot (1 + \lambda \sigma(h_\theta(X_i, S_i)))\right)$, with sensitivity $\lambda = 0.5$. Conditioning on $\sigma(h_\theta)$ models rational self-selection: individuals who perceive a higher approval probability are more likely to apply.

Action and observation. The agent outputs a continuous approval probability $a \in [0, 1]$; the realised decision is $\text{Bernoulli}(a)$. A stochastic action space rather than a hard binary decision allows the policy to express calibrated uncertainty and enables gradient-based training without discrete relaxations. The 12-dimensional observation contains the current applicant’s normalised wealth and group, group mean wealths (μ_M, μ_F) , Hawkes intensities (λ_R, λ_B) , cumulative default rates (ρ_M, ρ_F) , the approval model score $\sigma(h_\theta)$, learned coefficients (θ_S, θ_X) , and loan amount l_i . Bank revenue per decision: $+(1 + r_{\text{int}})l_i$ on repayment, $-l_i$ on default, 0 on rejection.

D.3 ALGORITHMIC DETAILS

D.3.1 HEURISTICS DETAILS

We evaluate seven rule-based policies, summarised in Table 4. The one-step predictor is the only policy with learnable structure; the remainder are parameter-free.

Table 4: Rule-based baseline policies.

Policy	Symbol	Decision rule	Parameters
Oracle	$\mathbb{O}$	$a = g_i$ (ground-truth label)	-
Always Approve	$\mathbb{A}_a$	$a = 1$	-
Always Reject	$\mathbb{A}_r$	$a = 0$	-
Uniform Acceptance	$\mathbb{A}_U$	$a \sim \mathcal{U}(0, 1)$	-
Rich Becomes Richer	$\mathbb{A}_{R\uparrow}$	$a = 1$ if $g_i^t = 1$, else $a \sim \mathcal{U}(0, 1)$	-
Reverse Rich Becomes Richer	$\mathbb{A}_{B\uparrow}$	$a = 1$ if $g_i^t = 0$, else $a \sim \mathcal{U}(0, 1)$	-
One-step Predictor	$\mathbb{A}_H$	$a = \sigma(h_\theta(X_i, g_i))$	θ_g, θ_X, b

One-step Predictor ($\mathbb{A}_H$). The approval decision is based on a logistic regression model h_θ fit on the population, which predicts each applicant’s likelihood of repayment from their wealth and group membership. Concretely, the approval score is $\pi(X, g) = \sigma(h_\theta(X, g)) = \sigma(\theta_g g + \theta_X X + b)$, where $\tilde{X}$ is standardised wealth and σ is the sigmoid function. Parameters (θ_g, θ_X, b) are estimated via `LogisticRegression(random_state=42, max_iter=1000)` with L2 regularisation at $C = 1.0$, fit on the LOANLY population with target $\mathbf{1}[\text{income}_i > \$50\text{K}]$. The model is trained once at simulation onset and does not update during deployment, making it a static learned heuristic rather than an adaptive policy.

D.3.2 RL ALGORITHMS

Policy Gradient (PG). Our Passive Reinforcement Learning (RL) implementation is based on REINFORCE (Williams, 1992) with several modifications suited to the continuous, performative setting. The policy is a Beta distribution $\pi_\theta(a | s) = \text{Beta}(\alpha_\theta(s), \beta_\theta(s))$ over the approval probability $a \in [0, 1]$, parameterised by a two-layer MLP with ReLU activations. Rewards are normalised online using Welford’s algorithm (running mean and variance across all steps and episodes), and per-episode returns are additionally standardised to zero mean and unit variance, acting as a self-normalising baseline. An entropy bonus encourages exploration. Gradients are clipped to unit norm before each update. For constrained objectives, Lagrange multipliers λ_{wealth} and $\lambda_{\text{approval}}$ are learnable parameters updated jointly via a separate Adam optimiser. Hyperparameters are listed in Table 5.

Table 5: PG hyperparameters.

Hyperparameter	Value	Description
Hidden dim	128	MLP hidden layer width
Layers	2	MLP depth (ReLU activations)
Policy distribution	Beta	Over $a \in [0, 1]$
Optimiser	Adam	Policy parameters
Learning rate η	10^{-3}	Policy Adam lr
Discount γ	0.99	Return discounting
Entropy coef	0.01	Exploration bonus weight
Grad clip norm	1.0	Per-update gradient clipping
λ lr	10^{-2}	Lagrange multiplier Adam lr
λ_{wealth} (init)	2.0	Initial wealth constraint weight
$\lambda_{\text{approval}}$ (init)	2.0	Initial approval rate constraint weight
Reward normalisation	Welford online	Running mean/variance across all steps

Performative Policy Gradient (PePG). Our Performative Reinforcement Learning (PERL) implementation utilises PePG (Perdomo et al., 2020), which extends REINFORCE by explicitly accounting for the dependence of the environment’s transition distribution on the policy. The gradient estimate augments the standard policy gradient with two additional terms: a *transition gradient* capturing how the policy shapes future Hawkes intensities and group mean wealth, and a *reward gradient* capturing how the policy directly influences the reward distribution. A performative replay buffer stores recent episode transitions to stabilise these gradient estimates. The policy architecture, optimiser, and constrained optimisation mechanism are identical to PG. Entropy regularisation is disabled by default ($\lambda_H = 0$), as the performative gradient terms already provide sufficient signal. Hyperparameters are listed in Table 6.

Table 6: PePG hyperparameters.

Hyperparameter	Value	Description
Hidden dim	128	MLP hidden layer width
Layers	2	MLP depth (ReLU activations)
Policy distribution	Beta	Over $a \in [0, 1]$
Optimiser	Adam	Policy parameters
Learning rate η	10^{-3}	Policy Adam lr
Discount γ	0.99	Return discounting
Entropy coef λ_H	0.0	Disabled by default
Grad clip norm	1.0	Per-update gradient clipping
λ lr	10^{-2}	Lagrange multiplier Adam lr
Buffer capacity	50	Performative replay buffer (episodes)
Warmup episodes	0	Episodes before performative gradients
Transition weight	1.0	Weight on $\nabla_{\theta} \log P^{\pi_{\theta}}$
Reward weight	1.0	Weight on $\nabla_{\theta} r^{\pi_{\theta}}$
Hawkes weight	1.0	Hawkes component of transition gradient
Wealth weight	1.0	Wealth component of transition gradient
$\alpha_R = \alpha_B$	0.3	Hawkes excitation (matched to env)
$\beta_R = \beta_B$	2.0	Hawkes decay (matched to env)

E EXPERIMENTAL RESULTS

E.1 DIAGNOSING **Eutopia** WITH HEURISTIC POLICIES

We provide the detailed analysis of the performance of rule-based baselines elaborated in Appendix D.3.1. We examine four macro metrics (Section 3.5) as depicted in Figure 8

Wealth dynamics. Figure 8a shows wealth gap trajectories that behave as expected under each policy. Rich Becomes Richer ($\mathbb{A}_{R\uparrow}$) produces a steadily growing positive wealth gap favouring males, increasing monotonically across all 500 episodes with no sign of saturation. Reverse Rich Becomes Richer ($\mathbb{A}_{B\uparrow}$) produces the mirror image: a symmetric negative wealth gap of comparable magnitude, confirming that the group-level dynamics are not hard-coded into the environment but arise from the per-decision feedback on wealth accumulation and subsequent application propensity.

Always Approve ($\mathbb{A}_a$) peaks around episode 100 before saturating, consistent with the loan gain parameter κ_i bounding how much wealth a single cohort can accumulate per episode once approval is universal. Always Reject ($\mathbb{A}_r$) remains at exactly zero throughout: with no loans issued, no wealth changes for any individual, and the wealth gap is precisely zero. This is the cleanest sanity check the environment offers, confirming that all observed dynamics in other policies arise from the decision feedback loop rather than from any initialisation artefact.

Oracle ($\mathbb{O}$) and Uniform Acceptance ($\mathbb{A}_U$) both maintain small, stable wealth gaps throughout, as both policies spread approvals across the population without strong group preference. The One-Step Predictor ($\mathbb{A}_H$) is the notable exception among the non-preferential policies: it produces a noticeably higher wealth gap than Uniform Acceptance, and one that grows over time rather than stabilising. This reflects a structural limitation of static deployed policies: fixed at initialisation, the predictor encodes the demographic correlations present at that moment, and these correlations are then amplified by the feedback loop as the wealth distribution shifts under the policy’s own decisions.

Inequality ratio. Figure 8b reinforces the wealth gap observations through the inequality ratio $\rho(t) = \mu_M / \mu_F$, which measures the relative rather than absolute disparity between groups. Rich Becomes Richer drives ρ sharply upward in the first 100 episodes, reaching values well above one before declining gradually. The decline reflects the fact that female applicants continue to receive probabilistic approvals under this policy, so their wealth does grow over time, only more slowly than male wealth. By episode 500, ρ remains elevated but is no longer increasing, suggesting the groups are approaching a biased steady state.

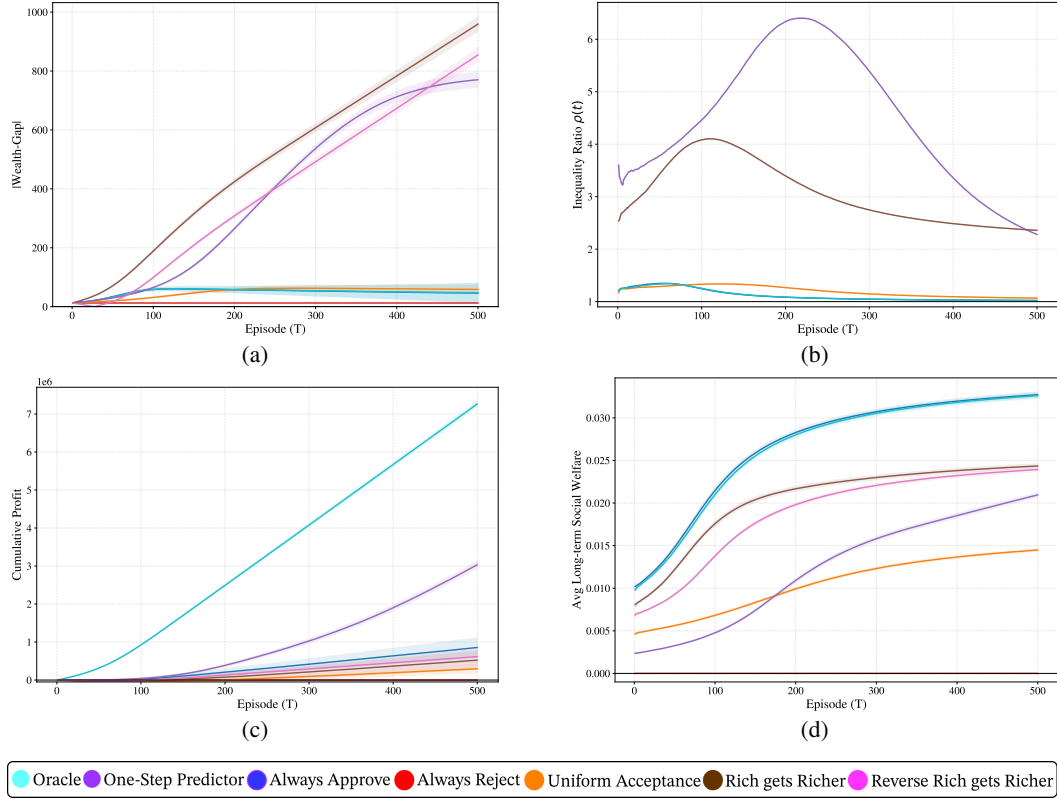


Figure 8: Evaluating Eutopia with Heuristic Algorithms

Reverse Rich Becomes Richer produces the mirror pattern, with ρ falling sharply below one before recovering slightly. The near-symmetry between the two trajectories is informative: it confirms that the Hawkes process parameters and initial wealth distributions do not introduce any hidden group-level asymmetry that would distort the feedback dynamics. All remaining policies cluster near $\rho = 1$ throughout, indicating that neither universal approval, random approval, nor a simple learned predictor introduces sustained group-level inequality in the long run, even if the absolute wealth gap for the One-Step Predictor grows slowly.

Profit. In Figure 8c, cumulative profit grows fastest under Always Approve, which accumulates roughly an order of magnitude more profit than any other policy by episode 500. This follows directly from the environment’s profit structure: since the net return on repayment exceeds the expected loss on default in expectation (Section D.2), approving every applicant is profitable in aggregate, and no selective policy can match it on raw revenue. Always Reject, as expected, accumulates zero profit throughout.

Rich Becomes Richer and Reverse Rich Becomes Richer achieve nearly identical cumulative profit despite directing their approvals at different groups. This confirms that demographic bias in either direction does not confer a financial advantage to the lender in this environment: both groups have similar aggregate repayment statistics by construction, and the Hawkes-driven application volumes are symmetric. Oracle provides a natural ceiling among the selective policies, outperforming the One-Step Predictor and Uniform Acceptance modestly but consistently. The gap between Oracle and the One-Step Predictor is small but persistent, consistent with the predictor’s mild tendency to approve lower-creditworthiness applicants in the wealthier group.

Long-term social welfare. Figure 8d shows per-episode welfare $\bar{R}$, which combines individual loan outcomes across both groups. Always Approve achieves the highest welfare alongside Oracle, with both saturating around episode 200. The saturation reflects the same bounding effect seen in the wealth gap: once individual wealth has grown to a level consistent with the loan parameters, the

marginal welfare gain per episode stabilises. Always Reject produces near-zero welfare throughout, since no loans are issued and no individuals benefit from the credit market.

The One-Step Predictor begins with lower welfare than Oracle but increases steadily across all 500 episodes, not yet saturating by the end of the deployment run. This persistent growth likely reflects the gradual accumulation of wealth among approved applicants, which improves repayment rates over time and thus raises the per-episode welfare delivered by the same static approval rule. Uniform Acceptance follows a similar trajectory but at a somewhat lower level, consistent with its lower selectivity compared to the predictor.

Taken together, these results confirm that [Eutopia](#) responds to policy structure in interpretable and consistent ways across all four metrics. The feedback dynamics are active and consequential: small differences in decision rules compound over episodes into meaningfully different long-term outcomes for wealth, inequality, profit, and welfare. This provides a reliable foundation for the reinforcement learning experiments that follow.

E.2 WHY LEARNING IS NEEDED: MOTIVATING RL

Rule-based policies offer interpretability but no adaptability: their decision rules are fixed at deployment and cannot respond to the distributional shifts they themselves induce. A static policy that performs well on the initial population may perform poorly after 100 episodes, precisely because its own decisions have reshaped the distribution it faces. To make this concrete, we compare Oracle and the One-Step Predictor against two fairness-constrained RL agents over 1000 episodes: a standard RL agent trained with a fairness Lagrangian (RL), and a performative RL agent (PERL) that additionally accounts for outcome fairness in the Lagrangian. The comparison is intentionally asymmetric: the static policies are fully specified at episode zero, while the learned agents continue to adapt during deployment. This is not a fair competition but a diagnostic—the goal is to understand what static policies cannot do, and where learned policies show promise. Figure 9 shows the four outcome metrics across all agents.

Wealth dynamics. Figure 9a plots the wealth gap $\mu_M - \mu_F$. The One-Step Predictor produces by far the largest sustained positive wealth gap: it grows rapidly between episodes 100 and 500, saturating near \$800K and remaining there through episode 1000. This is a direct consequence of the feedback loop. The predictor is fixed at initialisation, encodes the demographic correlations present at that moment, and then continues to act on those correlations even as the wealth distribution drifts under its own decisions. There is no mechanism by which it can detect or correct this drift. Oracle, by contrast, maintains a small and slowly declining wealth gap throughout, reflecting the fact that ground-truth creditworthiness labels are correlated with initial wealth but do not shift as wealth accumulates.

The two learned agents exhibit substantially wider variance across seeds in the early episodes, consistent with the stochastic exploration phase of online policy gradient updates. Their mean trajectories remain close to zero and well below Oracle throughout, and both show a slight upward trend in the later episodes that has not yet stabilised by episode 1000. The fairness Lagrangian gives the learned agents an explicit incentive to limit group-level wealth disparities, which the static policies entirely lack.

Inequality ratio. Figure 9b shows the inequality ratio $\rho(t)$. The One-Step Predictor begins around $\rho \approx 3.5$, rises to a peak of approximately 6.5 near episode 200, and then declines slowly, reaching roughly 1.3 by episode 1000. That it eventually declines is consistent with the wealth gap panel: the wealthier male group eventually saturates its loan-driven wealth gains, and the relative gap narrows even under a biased static policy. Importantly, ρ remains well above one at episode 1000, meaning the One-Step Predictor has not converged to a fair outcome within this deployment horizon.

Oracle tracks close to $\rho = 1$ throughout, as expected. PERL and RL begin with wide credible intervals during early deployment, but their mean ρ trajectories converge to approximately one by episode 600 and remain there. This convergence is the fairness Lagrangian working as intended: without it, a policy optimising only cumulative profit would have no incentive to maintain $\rho \approx 1$.

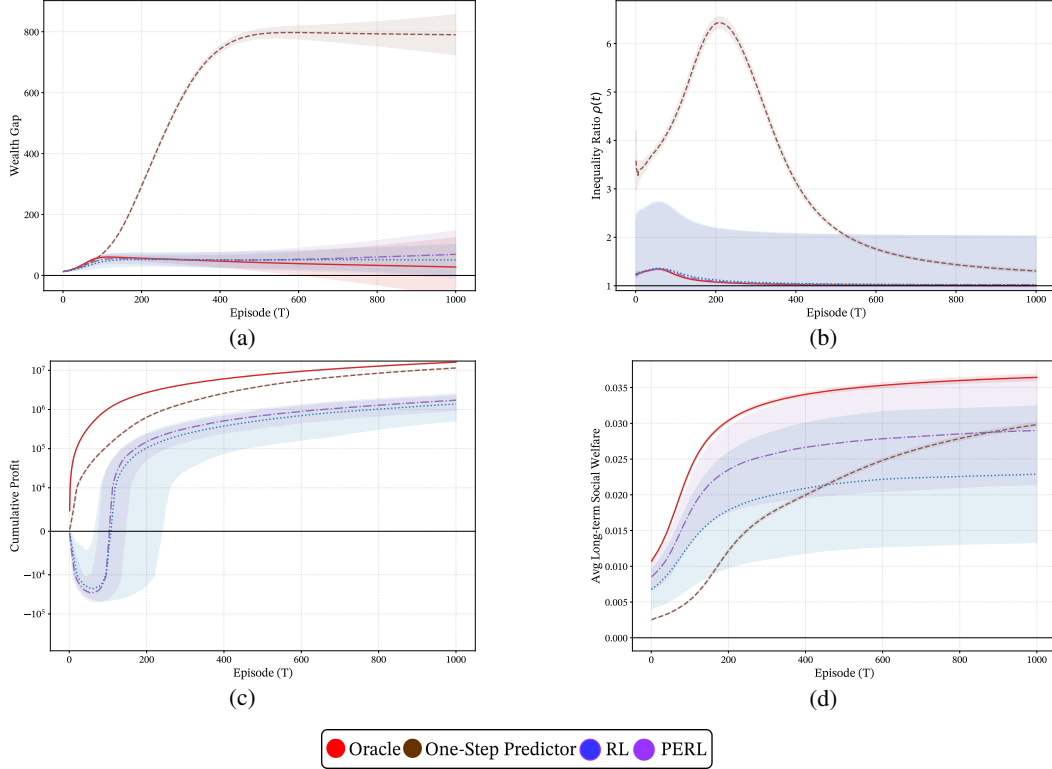


Figure 9: Learning vs. Heuristics. Expanded results from Figure 3. Learned agents (RL, PERL) adapt to the distributional shifts they induce, maintaining near-zero wealth gaps and balanced inequality ratios throughout deployment. Heuristic policies, while immediately profitable, encode the demographic correlations present at initialisation and have no mechanism to correct the wealth disparities that compound over time.

Profit. Figure 9c plots cumulative profit on a log scale, which is necessary to make the early-deployment behaviour of the learned agents visible alongside the steadily accumulating profit of the static policies. Oracle and the One-Step Predictor accumulate profit from episode one, as their deterministic decision rules do not include any exploratory decisions that would result in unprofitable approvals. Oracle leads throughout, with the One-Step Predictor approximately half an order of magnitude behind by episode 1000.

Both RL and PERL begin with negative cumulative profit, dipping to around -10^4 near episode 100 before recovering. This is the cost of early exploration: in the initial deployment phase, the policy is still adapting, and the resulting defaults can outweigh repayments. As the policy improves, cumulative profit turns positive and continues to grow, reaching approximately 10^6 by episode 1000 for both agents. Critically, neither RL nor PERL has saturated its profit trajectory by episode 1000, whereas Oracle and the One-Step Predictor have both largely plateaued. The implication is that learned policies, given a sufficient deployment horizon, are on a trajectory to close the gap with static rules.

Long-term social welfare. Figure 9d shows per-episode welfare $\bar{R}$. Oracle achieves the highest welfare throughout, reaching approximately 0.036 by episode 1000 and beginning to show signs of saturation. PERL performs second best, reaching approximately 0.029 by episode 1000 and still growing, while the One-Step Predictor reaches a similar level but with considerably wider variance. RL lags behind both, reaching approximately 0.023 by episode 1000. Notably, none of the learned agents have saturated by episode 1000, in contrast to Oracle which is visibly flattening. This persistent growth indicates that the learned policies are still finding more welfare-efficient approval strategies over the deployment horizon, and points to the need for longer deployment runs in the full experiments.

Taken together, these results make the case for learned policies in performative environments. Static policies are immediately interpretable and profitable, but they are blind to the distributional conse-

Table 7: **RL: Ranking of fairness objectives across fairness perspectives based on long-term performance.** Each group of columns reports the best observed fairness objective and its mean $\pm$ standard deviation across seeds.

Fairness	Cumulative Profit($\times 10^7$)		Wealth Gap		Social Welfare $\bar{R}$		Inequality ρ	
	Best	Mean $\pm$ Std	Best	Mean $\pm$ Std	Best	Mean $\pm$ Std	Best	Mean $\pm$ Std
Outcome	FL	138.420 $\pm$ 86.722	RMM	49.050 $\pm$ 45.353	FL	0.023 $\pm$ 0.013	FL	0.907 $\pm$ 0.261
DM	UP	116.322 $\pm$ 107.624	RMM	57.980 $\pm$ 25.737	UP	0.024 $\pm$ 0.014	RMM	0.936 $\pm$ 0.259
Two-Sided	UP	166.211 $\pm$ 72.246	UP	29.559 $\pm$ 66.819	FL	0.030 $\pm$ 0.005	RMM	0.991 $\pm$ 0.051

Table 8: **PERL: Ranking of fairness objectives across fairness perspectives based on long-term performance.** Each group of columns reports the best observed fairness objective and its mean $\pm$ standard deviation across seeds.

Fairness	Cumulative Profit($\times 10^7$)		Wealth Gap		Social Welfare $\bar{R}$		Inequality ρ	
	Best	Mean $\pm$ Std	Best	Mean $\pm$ Std	Best	Mean $\pm$ Std	Best	Mean $\pm$ Std
Outcome	FL	174.408 $\pm$ 74.473	RMM	47.734 $\pm$ 64.911	FL	0.029 $\pm$ 0.010	FL	1.018 $\pm$ 0.049
DM	UP	112.483 $\pm$ 83.210	UP	54.948 $\pm$ 27.794	FL	0.019 $\pm$ 0.008	UP	0.962 $\pm$ 0.130
Two-Sided	RMM	123.561 $\pm$ 90.776	RMM	34.017 $\pm$ 71.174	SW	0.022 $\pm$ 0.014	UP	0.874 $\pm$ 0.291

quences of their own decisions: the One-Step Predictor’s wealth gap grows to nearly \$800 and its inequality ratio peaks at 6.5, precisely because it has no mechanism to respond to the shift it induces. Learned policies incur a real upfront cost in profit during early deployment, but their trajectories in profit and welfare show no saturation by episode 1000, whereas the static policies have plateaued well before that. The remainder of this section examines whether, and under which fairness objectives, the learned policies can be trained to reliably outperform the static baselines across all four metrics simultaneously.

E.3 PASSIVE VS. PERFORMATIVE MODELS OF RL

The previous section established that learned policies outperform static rules in the long run, but treated fairness as a single constraint. In practice, what it means for a lending policy to be fair is not self-evident: different stakeholders have different and sometimes conflicting notions of fairness, and the choice of fairness objective shapes not only the policy’s behaviour but also the long-term distributional consequences it produces. We consider three distinct fairness objectives drawn from the framework in Section 3.4: DM fairness, which constrains the approval rate disparity across groups and reflects the decision-maker’s institutional interest in consistent treatment; outcome fairness, which constrains the wealth gap between groups and reflects a societal interest in equitable long-term outcomes; and two-sided fairness, which interpolates between the two via the parameter α and captures the tension between these perspectives.

Each objective enters the procedure as a Lagrangian penalty on the policy gradient update, so the agent learns to maximise cumulative profit subject to the chosen fairness constraint. We further ask whether the agent’s model of the environment matters: a passive RL agent treats each episode as drawn from a fixed distribution and ignores the feedback loop, while a performative RL agent (PERL) explicitly accounts for the fact that its decisions reshape the distribution it will face in the next episode. For each fairness objective, we deploy both a passive RL agent and a PERL agent and compare their trajectories across all four outcome metrics. The goal is to understand which fairness objectives are compatible with long-term profit and welfare, and whether modelling the performative feedback loop leads to meaningfully different outcomes than ignoring it.

E.3.1 GAIN: PERFORMANCE OF BEST ALGORITHMS

Tables 7, 8 report, for each fairness objective, the best-performing fairness perspective on each metric and its observed value at the end of deployment, for the passive (RL) and performative (PERL) agents respectively. Alternatively, Tables 9 and 10 depict for each fairness objective, the best choice of fairness perspective at the end of deployment.

Table 9: **RL: Ranking of fairness perspectives across fairness objectives based on long-term performance.** Each group of columns reports the best observed fairness perspective and its mean $\pm$ standard deviation across seeds.

Reward	Cumulative Profit($\times 10^7$)		Wealth Gap		Social Welfare $\bar{R}$		Inequality ρ	
	Best	Mean $\pm$ Std	Best	Mean $\pm$ Std	Best	Mean $\pm$ Std	Best	Mean $\pm$ Std
UP	Two-S.	166.211 $\pm$ 72.246	Two-S.	29.559 $\pm$ 66.819	Two-S.	0.028 $\pm$ 0.007	Two-S.	0.996 $\pm$ 0.054
SW	Two-S.	164.898 $\pm$ 88.075	Two-S.	51.991 $\pm$ 72.339	Two-S.	0.027 $\pm$ 0.010	Outcome	0.978 $\pm$ 0.061
RMM	Outcome	93.306 $\pm$ 82.358	Two-S.	48.777 $\pm$ 31.473	Two-S.	0.017 $\pm$ 0.003	DM	0.936 $\pm$ 0.259
FL	Two-S.	147.142 $\pm$ 83.210	Two-S.	35.822 $\pm$ 78.928	Two-S.	0.030 $\pm$ 0.005	Outcome	0.907 $\pm$ 0.261

Table 10: **PERL: Ranking of fairness perspectives across fairness objectives based on long-term performance.** Each group of columns reports the best observed fairness perspective and its mean $\pm$ standard deviation across seeds.

Reward	Cumulative Profit($\times 10^7$)		Wealth Gap		Social Welfare $\bar{R}$		Inequality ρ	
	Best	Mean $\pm$ Std	Best	Mean $\pm$ Std	Best	Mean $\pm$ Std	Best	Mean $\pm$ Std
UP	DM	112.483 $\pm$ 83.210	Two-S.	36.084 $\pm$ 26.702	Two-S.	0.017 $\pm$ 0.015	Two-S.	0.874 $\pm$ 0.291
SW	Outcome	140.769 $\pm$ 101.080	Two-S.	35.610 $\pm$ 70.530	Outcome	0.024 $\pm$ 0.013	Outcome	1.056 $\pm$ 0.191
RMM	Two-S.	123.561 $\pm$ 90.776	Two-S.	34.017 $\pm$ 71.174	Two-S.	0.020 $\pm$ 0.014	DM	0.990 $\pm$ 0.064
FL	Outcome	174.408 $\pm$ 74.473	Two-S.	50.442 $\pm$ 33.648	Outcome	0.029 $\pm$ 0.010	DM	1.016 $\pm$ 0.038

Outcome fairness. Under outcome fairness, FL achieves the best cumulative profit for both RL (138.420 $\pm$ 86.722) and PERL (174.408 $\pm$ 74.473), with PERL’s advantage amounting to approximately \$0.36B simulation units and a notably lower standard deviation. This reduction in variance alongside the gain in mean performance suggests that modelling the performative feedback loop not only improves expected profit but stabilises it across seeds. FL also achieves the best social welfare for both agents (0.023 $\pm$ 0.013 for RL and 0.029 $\pm$ 0.010 for PERL), again with PERL leading by a consistent margin. The pattern across fairness objectives is notably uniform: no single fairness objective dominates the others by a large margin under outcome fairness for either agent, indicating that the fairness perspective itself is the primary driver of performance in this regime rather than the specific reward shaping.

RMM achieves the smallest wealth gap for both RL (49.050 $\pm$ 45.353) and PERL (47.734 $\pm$ 64.911), with comparable means but substantially higher variance for PERL. The inequality ratio tells a sharper story: PERL achieves $\rho = 1.018 \pm 0.049$ under FL, essentially at parity, while RL achieves only 0.907 $\pm$ 0.261 under the same fairness objective. The difference in both the mean and the variance points to a qualitative distinction: the performative agent, by anticipating the distributional consequences of its decisions, not only achieves better group-level balance on average but does so more reliably across seeds.

DM fairness. Under DM fairness, both agents prefer UP for cumulative profit, achieving similar levels (116.322 $\pm$ 107.624 for RL and 112.483 $\pm$ 83.210 for PERL). The high standard deviations relative to the means in both cases reflect substantial seed-level variability, and the two agents are effectively indistinguishable in profit under this constraint. This suggests that constraining approval rate disparity does not create a strong signal that differentiates either agent type or fairness objective in terms of profit.

The agents diverge more clearly on welfare and wealth gap. RL’s best welfare is achieved by UP (0.024 $\pm$ 0.014), while PERL’s is achieved by FL (0.019 $\pm$ 0.008), a reversal from the outcome fairness results and a lower absolute value for PERL despite its generally stronger performance elsewhere. The wealth gap is the largest of any fairness objective for both agents: RMM achieves 57.980 $\pm$ 25.737 for RL and UP achieves 54.948 $\pm$ 27.794 for PERL, both substantially higher than under two-sided or outcome fairness. This is a structural consequence of DM fairness: constraining approval rate disparity does not penalise wealth divergence, and the two groups continue to drift apart in accumulated wealth regardless of how the fairness objective is shaped or whether the agent models the feedback loop. This finding is consistent with the radar plots, where the DM polygon is nearly collapsed on the wealth gap axis.

Two-sided fairness. Two-sided fairness produces the strongest results for RL. UP achieves the highest profit of any RL configuration across all three fairness objectives (166.211 ± 72.246), and FL achieves the best social welfare (0.030 ± 0.005), with the smallest standard deviation of any welfare result in either table, indicating that FL consistently achieves high welfare under this constraint regardless of seed. RMM achieves the best inequality ratio (0.990 ± 0.051), the closest to parity of any RL configuration, and the best wealth gap (29.559 ± 66.819), though the high variance on the wealth gap suggests that this result is not reliable across all seeds.

PERL performs more modestly under two-sided fairness. RMM achieves the best profit (123.561 ± 90.776) and wealth gap (34.017 ± 71.174), both with wide standard deviations, and no metric shows a clear separation across fairness objective. This stands in contrast to RL’s behaviour under the same objective and to PERL’s own behaviour under outcome fairness, suggesting that the performative gradient interacts less effectively with the interpolated fairness signal of two-sided fairness. The radar plots corroborate this: the two-sided polygon for PERL is the most spread of any constraint, indicating that the reward function choice has the most consequential and least predictable effect on the performance profile under this objective.

Effect of fairness objective on fairness perspective selection. Table 3 shows which fairness perspective each fairness objective most benefits from. For RL, two-sided fairness dominates across fairness objectives: it is the best fairness perspective for UP, SW, and FL on both profit and welfare, with RMM being the only exception where outcome fairness achieves the best profit (93.306 ± 82.358). For PERL, outcome fairness is more consistently preferred: FL under outcome fairness achieves the strongest results in the entire PERL table, with the best profit (174.408 ± 74.473) and welfare (0.029 ± 0.010). For UP and RMM under PERL, preferences split between DM and two-sided fairness across metrics, with no consistent pattern, reflecting the weaker signal these fairness perspective provide to the performative gradient. A notable cross-agent observation is that for every fairness objective, PERL and RL select different best fairness perspective on at least one metric, confirming that the two agents respond to fairness constraints in qualitatively different ways rather than simply scaling the same behaviour.

E.3.2 EVOLUTION OF PERFORMANCES

Outcome Fairness. Figures 11 and 10 show results under outcome fairness for RL and PERL respectively. Under this objective, the fairness objective target group wealth directly: SW maximises the sum of group mean wealths, RMM maximises the minimum group mean wealth, and FL penalises the absolute difference in group mean wealth. Because the fairness constraint and the wealth gap metric are now measuring related quantities, the dynamics here are more directly interpretable than under DM fairness.

Wealth gap. Under RL (Figure 11a), all three fairness objective stabilise relatively quickly: FL and RMM converge tightly near \$50K by episode 200 and remain there, while SW settles slightly higher around \$62K with wider variance. That RMM achieves the smallest wealth gap is consistent with its maximin structure, which directly penalises any imbalance between group wealths. Under PERL (Figure 10a), the picture is different: FL and SW stabilise around \$55K at episode 200 but then continue growing slowly, with FL reaching approximately \$70K by episode 1000. RMM remains the lowest at around \$51K and is more stable, though all three credible intervals are wide. The continued growth of FL under PERL despite an explicit wealth gap penalty in the reward suggests that the performative gradient is trading off some long-term wealth parity for improved profit and welfare, as seen in the other panels.

Inequality ratio. Under RL (Figure 11b), all three fairness objectives peak near episode 80–100 at $\rho \approx 1.30$ – 1.35 , with FL peaking highest. All three then decline, with SW descending the fastest and reaching $\rho \approx 1.03$ by episode 1000, the closest to parity of the three. FL and RMM both end near $\rho \approx 1.07$, with overlapping credible intervals. Under PERL (Figure 10b), the contrast between fairness objectives is sharper. FL and SW decline steeply after their peak, converging to $\rho \approx 1.05$ by episode 1000 with credible intervals that touch $\rho = 1$. RMM, however, declines considerably more slowly, still sitting near $\rho \approx 1.12$ at episode 1000 with a wide interval. This is the opposite of the wealth gap result, where RMM produced the smallest gap: under PERL, RMM’s maximin structure appears to concentrate its fairness effort on the wealth level of the weaker group rather than the ratio between groups, creating a tension between the two fairness metrics over the deployment horizon.

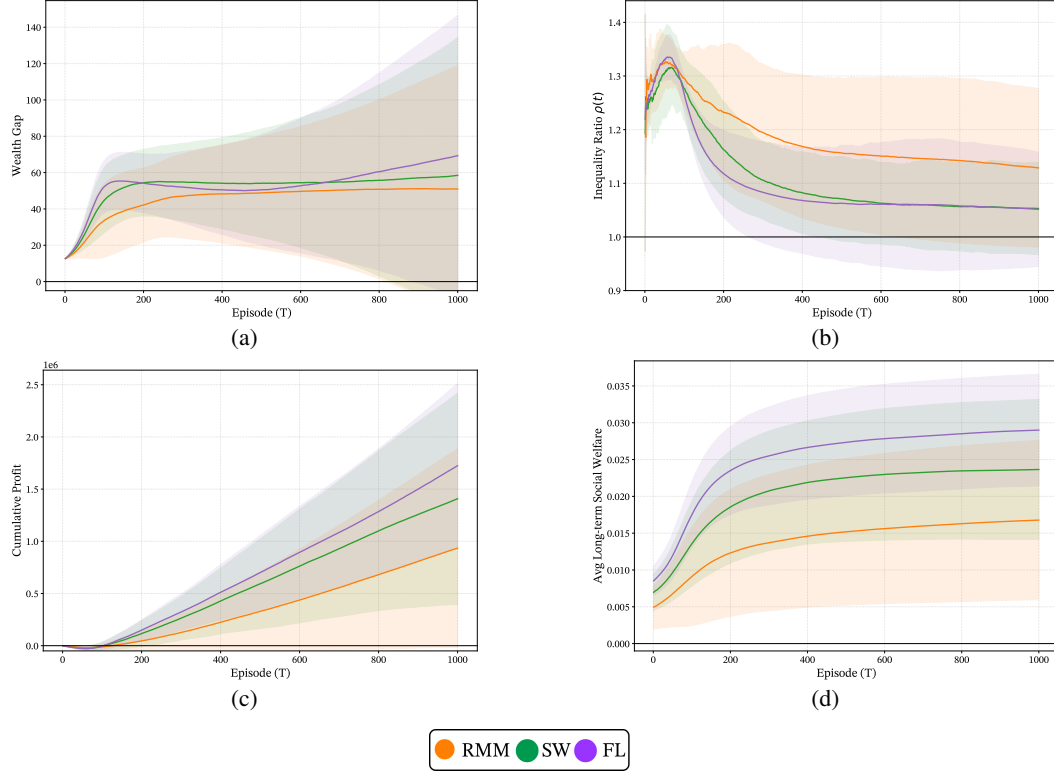


Figure 10: **PeRL: Outcome Fairness.** Long-term performance of PERL across Fairness Objectives under outcome fairness, tracked over $T = 1000$ episodes. Mean $\pm$ std across seeds.

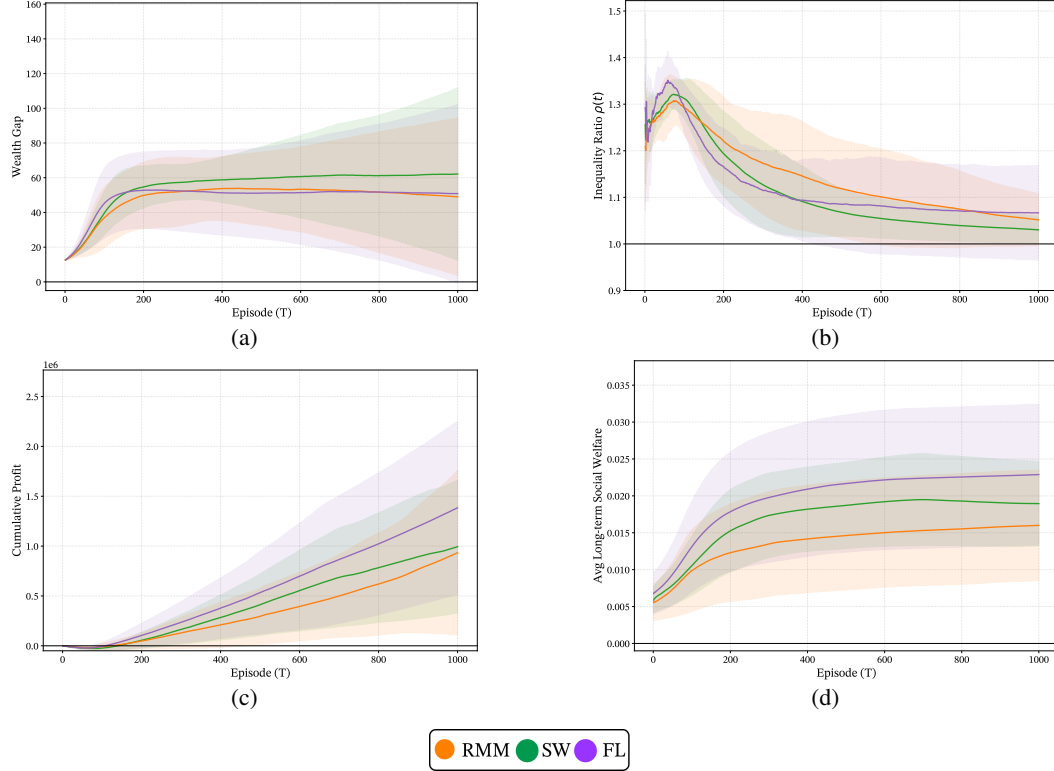


Figure 11: **RL: Outcome Fairness.** Long-term performance of RL across Fairness Objective under outcome fairness, tracked over $T = 1000$ episodes. Mean $\pm$ std across seeds.

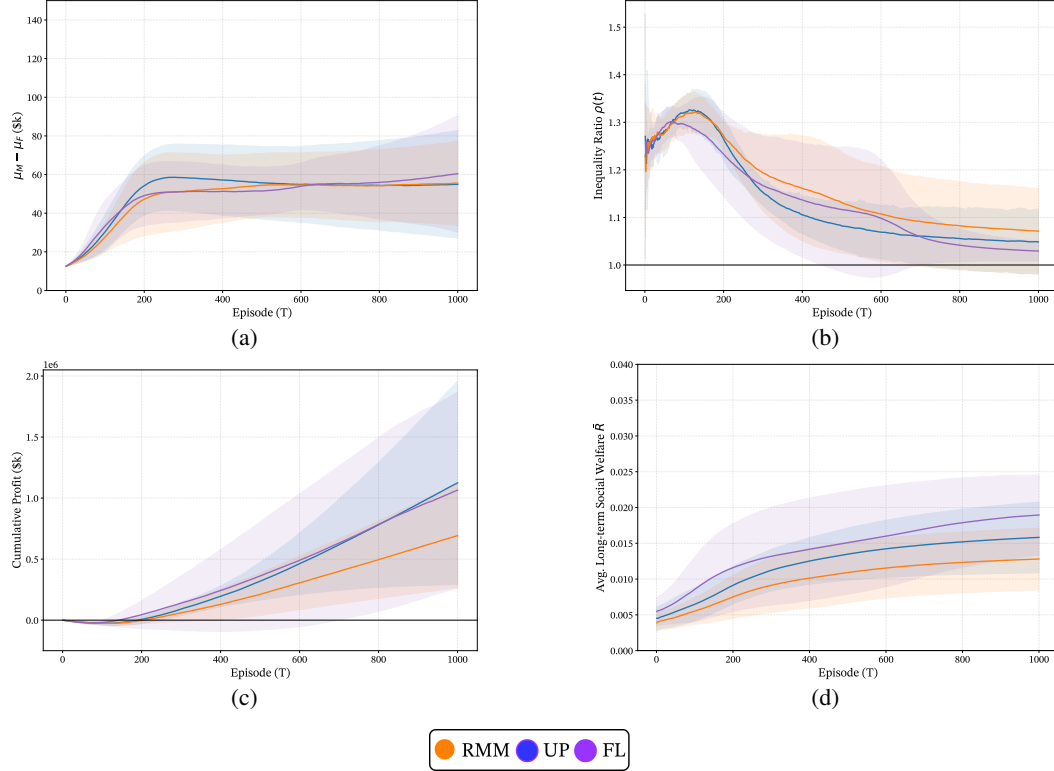


Figure 12: **PeRL: DM's Fairness.** Long-term performance of PERL across Fairness Objectives under DM's fairness, tracked over $T = 1000$ episodes. Mean $\pm$ std across seeds.

Cumulative profit. Under RL (Figure 11c), FL leads clearly and grows linearly throughout, reaching approximately \$1.38B by episode 1000. SW follows at approximately \$1.0B and RMM at \$0.93B. None of the three have reached a plateau by episode 1000, indicating that all fairness objectives are still improving their policies within this deployment horizon. Under PERL (Figure 10c), the ordering is the same but the gaps are larger: FL reaches approximately \$1.73B, SW approximately \$1.4B, and RMM approximately \$0.93B. The fact that PERL with FL achieves substantially higher profit than RL with FL, while RMM achieves identical profit under both agents, suggests that the performative gradient interacts most productively with the Lagrangian reward structure and provides the least additional benefit when the reward already has a hard maximin constraint.

Social welfare. Under RL (Figure 11d), FL leads at $\bar{R} \approx 0.023$ by episode 1000, followed by SW at 0.019 and RMM at 0.016. All three are still growing and show no sign of saturation. The ordering matches the profit panel exactly, suggesting that under outcome fairness for the passive agent, the reward function that achieves the highest profit also achieves the highest welfare, with no visible trade-off between the two over the deployment horizon. Under PERL (Figure 10d), this same ordering holds but the absolute values are higher across the board: FL reaches $\bar{R} \approx 0.029$, SW approximately 0.023, and RMM approximately 0.017. The welfare gap between FL and RMM is larger under PERL than under RL, again pointing to the Lagrangian reward as the primary beneficiary of performative modelling under outcome fairness.

DM Fairness. Figures 13 and 12 show results under DM fairness for RL and PERL respectively. Under this objective, the reward functions target per-step profit disparity between groups: FL penalises the absolute difference in per-step bank profit across groups, RMM maximises the minimum per-step profit across groups, and UP optimises aggregate profit with no fairness term. Because none of these objectives directly target wealth divergence, the two groups can drift apart in wealth even as their per-step profit contributions are equalised, and the dynamics here are structurally less interpretable than under outcome fairness.

Wealth gap. Under RL (Figure 13a), the wealth gap grows for all fairness objectives throughout the full 1000 episodes with no sign of stabilisation. UP produces the largest and most variable gap, reaching approximately \$80K by episode 1000 with a credible interval spanning nearly the

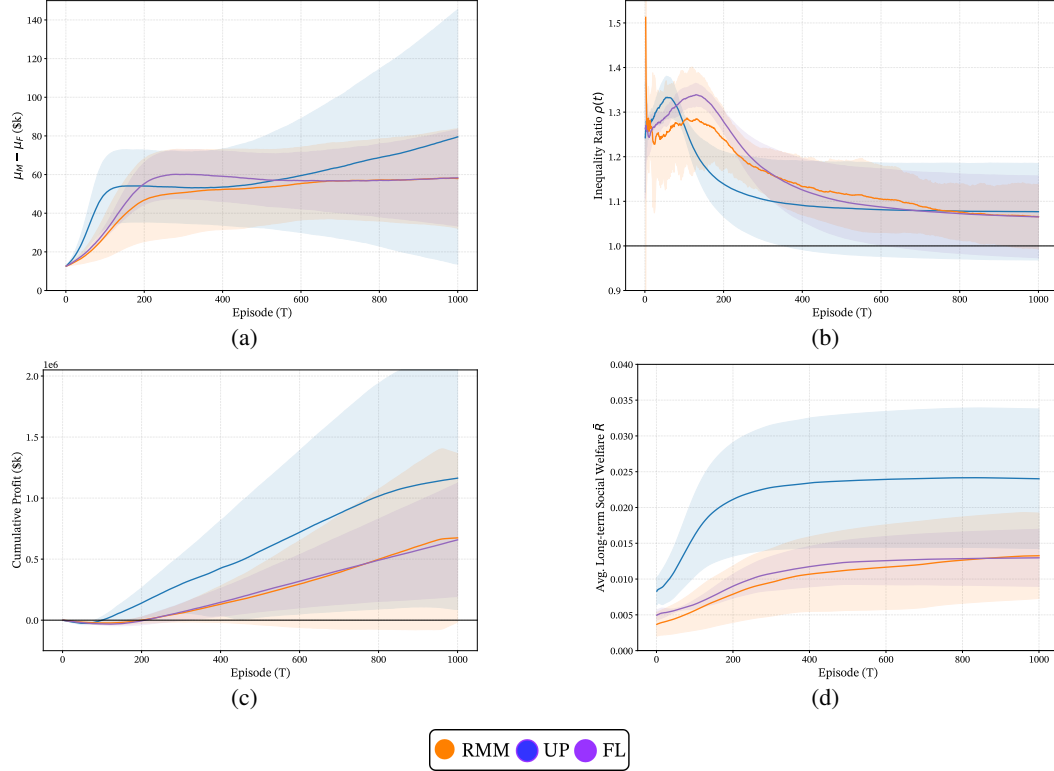


Figure 13: **RL: DM's Fairness.** Long-term performance of RL across Fairness Objectives under DM's fairness, tracked over $T = 1000$ episodes. Mean $\pm$ std across seeds.

full range of the plot. FL and RMM converge to a tighter cluster around \$58K–\$60K by episode 200 and remain there, suggesting that penalising profit disparity or maximising the group minimum introduces an implicit regularisation on wealth divergence even though neither objective directly targets it. Under PERL (Figure 12a), the three fairness objectives behave much more similarly: they all peak around \$55K–\$75K near episode 200 before stabilising, with heavily overlapping credible intervals. The performative gradient appears to equalise behaviour across reward functions on this metric, substantially reducing the sensitivity to fairness objectives choice that was visible for RL.

Inequality ratio. Under RL (Figure 13b), all fairness objectives begin near $\rho \approx 1.25$ –1.3. RMM spikes sharply in the first few episodes, briefly reaching $\rho \approx 1.5$, before declining. This early spike is consistent with RMM's maximin structure: in early deployment, maximising the minimum group profit can temporarily concentrate lending on the higher-wealth group before the policy stabilises. FL peaks later around episode 100 at approximately 1.33, while UP peaks around 1.3 and then declines fastest of the three, reaching $\rho \approx 1.07$ by episode 1000. All three converge near $\rho \approx 1.07$ –1.08 by episode 1000, still meaningfully above parity. Under PERL (Figure 12b), the early spike disappears entirely and the trajectories are considerably smoother throughout. UP and RMM peak together near $\rho \approx 1.32$ –1.35 around episode 150, while FL peaks slightly lower and declines faster. All three converge toward $\rho \approx 1.05$ –1.08 by episode 1000, with FL tracking closest to parity in the later episodes.

Cumulative profit. Under RL (Figure 13c), all fairness objectives begin with briefly negative cumulative profit before turning positive around episode 100–150. UP then separates sharply, reaching approximately \$1.15B by episode 1000 with a wide credible interval, while FL and RMM cluster together near \$650–\$670M. The persistent gap between UP and the other two reflects the direct profit cost of both the Lagrangian penalty in FL and the maximin structure of RMM under a constraint that already limits approval rate disparity. Under PERL (Figure 12c), the ordering changes: UP and FL become nearly indistinguishable, both reaching approximately \$1.1B by episode 1000, while RMM lags at approximately \$700M. The convergence of UP and FL under PERL is notable: once the agent anticipates the feedback loop, the Lagrangian penalty in FL becomes less binding, and the two fairness objectives discover policies of comparable profitability over the deployment horizon.

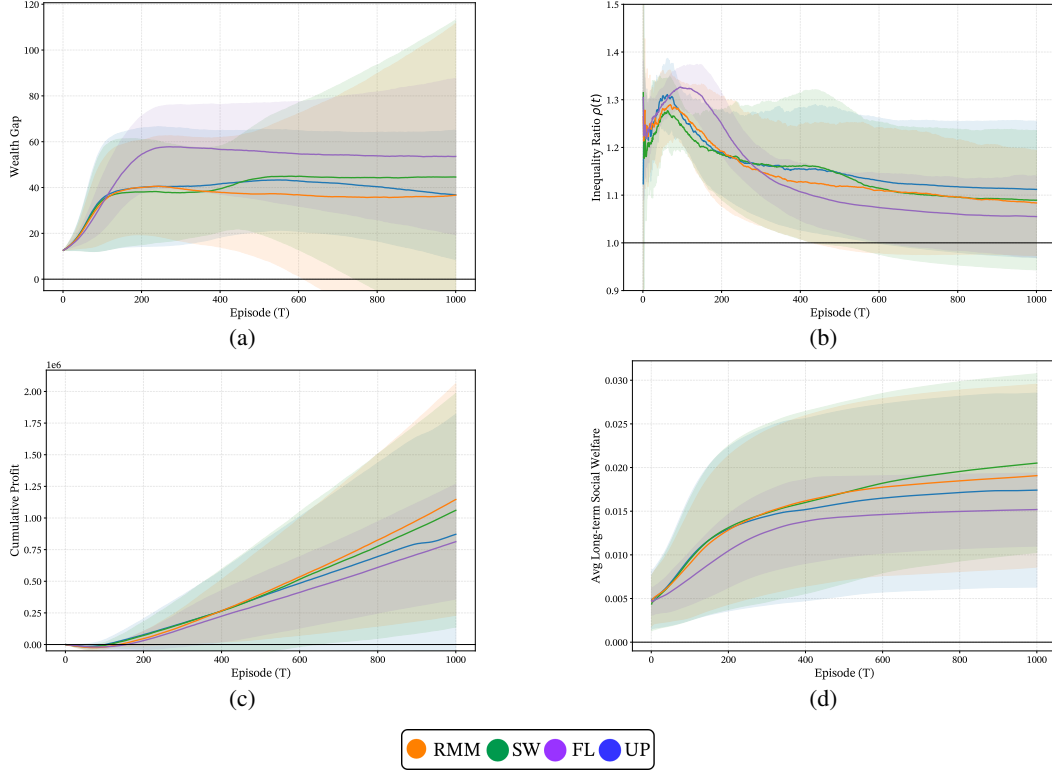


Figure 14: **PeRL: Two-Sided Fairness**. Long-term performance of PERL across fairness objectives under Two-Sided fairness, tracked over $T = 1000$ episodes. Mean $\pm$ std across seeds.

Social welfare. Under RL (Figure 13d), UP leads clearly at $\bar{R} \approx 0.024$, beginning to saturate around episode 400–600, while FL and RMM cluster around $\bar{R} = 0.013$ and are still slowly increasing at episode 1000. The gap between UP and the other two is the largest of any metric under DM fairness: UP’s more aggressive approval strategy generates more loans and thus more welfare per episode, at the cost of the largest wealth gap. Under PERL (Figure 12d), the ordering reverses: FL leads at approximately $\bar{R} = 0.019$, followed by UP at 0.016 and RMM at 0.013, and none of the three have reached a plateau by episode 1000. The welfare ordering reversal between UP under RL and FL under PERL points to a consistent pattern across both DM and outcome fairness: the Lagrangian reward becomes welfare-enhancing when the agent models the performative consequences of its decisions, while the unconstrained profit reward loses its welfare advantage once the feedback loop is explicitly accounted for.

Two-Sided Fairness. Figures 15 and 14 show results under two-sided fairness, which interpolates between outcome and DM fairness via the parameter α . Because the constraint combines both approval rate disparity and wealth gap penalisation, the dynamics here are richer than under either objective alone, and the behaviour of the two agents diverges most sharply of the three regimes considered.

Wealth gap. Under RL (Figure 15a), all four fairness objectives peak near episode 100–150 at approximately \$60K before diverging. UP and FL decline steadily throughout the remainder of the deployment run, reaching approximately \$30K and \$35K respectively by episode 1000. This is a qualitatively different behaviour from DM and outcome fairness, where wealth gaps either grew or stabilised: here, the combined constraint is actively pushing the wealth gap down. SW declines more slowly, remaining near \$50K by episode 1000, while RMM peaks later and settles around \$51K. Under PERL (Figure 14a), the picture is more stable but also more separated: FL stabilises around \$53K–\$54K after episode 200 and barely moves. SW and UP cluster around \$44K–\$45K, and RMM achieves the smallest gap at approximately \$37K. The declining trajectory seen for UP and FL under RL disappears under PERL, replaced by stable but differentiated levels.

Inequality ratio. Under RL (Figure 15b), all four fairness objectives peak between episodes 50 and 80 at $\rho \approx 1.35$ – 1.38 , with UP peaking highest, and then decline sharply and tightly. By episode

1000, all four converge near $\rho \approx 1.02$, with credible intervals touching one. RMM lags slightly throughout but reaches the same endpoint. This is the tightest convergence to parity of any RL configuration across all three fairness perspectives, and it occurs for all four fairness objectives simultaneously, suggesting that the two-sided constraint is effective at driving the inequality ratio toward one regardless of fairness objective choice over the deployment horizon. Under PERL (Figure 14b), convergence is slower and the fairness objectives separate more clearly. FL declines the most, reaching $\rho \approx 1.05$ by episode 1000. RMM and UP track together near $\rho \approx 1.09$, while UP stabilises around $\rho \approx 1.11$. SW has the widest credible interval of any curve, with the lower bound dipping below one, indicating that some seeds over-correct and temporarily favour the initially weaker group.

Cumulative profit. Under RL (Figure 15c), UP, SW, and FL cluster closely together, all growing linearly and reaching approximately \$1.47B–\$1.65B by episode 1000. RMM lags considerably at approximately \$0.88B, a gap that opens gradually from episode 200 onward and does not close by episode 1000. Under PERL (Figure 14c), the ordering reverses entirely: RMM now leads at approximately \$1.13B, followed by SW at \$1.06B, while UP and FL trail at \$0.84B and \$0.82B respectively. This reversal is the most striking cross-agent difference in the two-sided regime and mirrors the finding from outcome fairness, where PERL shifted the profit advantage toward the more constrained fairness objectives.

Social welfare. Under RL (Figure 15d), UP leads early, reaching approximately $\bar{R} = 0.028$ around episode 300, but then plateaus. FL rises more slowly and overtakes UP around episode 700, continuing to grow to approximately 0.030 by episode 1000. SW ends near 0.027. RMM is well separated from the other three throughout, reaching only approximately 0.016 and still growing at episode 1000. Under PERL (Figure 14d), SW leads at approximately $\bar{R} = 0.020$, followed closely by RMM at 0.019 and UP at 0.017. FL trails at approximately 0.015 throughout, a reversal from its leading position under RL. None of the four have reached a plateau by episode 1000. The swap in welfare leadership between FL under RL and SW under PERL, alongside the profit leadership moving from UP to RMM, points to a consistent pattern: under two-sided fairness, PERL redistributes performance away from the Lagrangian reward and toward the structure-based rewards, while RL does the opposite.

E.3.3 MACRO-ANALYSIS: PARETO FRONTIERS OF UTILITIES

Figure 16 presents radar plots summarising the long-term performance profile of PERL under each fairness constraint. We focus on PERL throughout this analysis, as it explicitly models the performative feedback loop and is the subject of the micro-analysis that follows. Each plot shows four normalised axes, all oriented so that a larger value is better: social welfare $\bar{R}$, cumulative profit, wealth gap plotted as its reciprocal so that a larger value corresponds to a smaller gap, and equality measured as proximity to $\rho = 1$, i.e. $|\rho - 1|$ inverted so that a larger value indicates a more equal outcome. The axes are normalised to a common visual scale to enable shape comparison across constraints. A larger and more symmetric polygon indicates better and more balanced performance across all four dimensions simultaneously. Each polygon corresponds to a different fairness perspective.

DM fairness. Figure 16a shows a pronounced kite shape elongated toward profit and equality. DM fairness achieves strong performance on both of these axes: the policy learns to maximise aggregate lending revenue while keeping the inequality ratio close to one, which is precisely what the approval rate disparity constraint incentivises. However the polygon is nearly collapsed on the wealth gap axis, confirming the structural limitation of DM fairness established in the aggregate analysis: constraining per-step profit disparity between groups does not prevent the two groups from diverging in accumulated wealth over the deployment horizon. Social welfare occupies an intermediate position. Crucially, all fairness objectives produce nearly identical polygon shapes under DM fairness, indicating that the constraint dominates the fairness objectives in determining the performance profile and that fairness objective choice has little leverage over long-term outcomes in this regime.

Outcome fairness. Figure 16b shows the most balanced and symmetric diamond of the three constraints. All four axes receive meaningful coverage: social welfare and profit are both strong, equality is moderate, and while the wealth gap axis remains the weakest of the four, it is noticeably

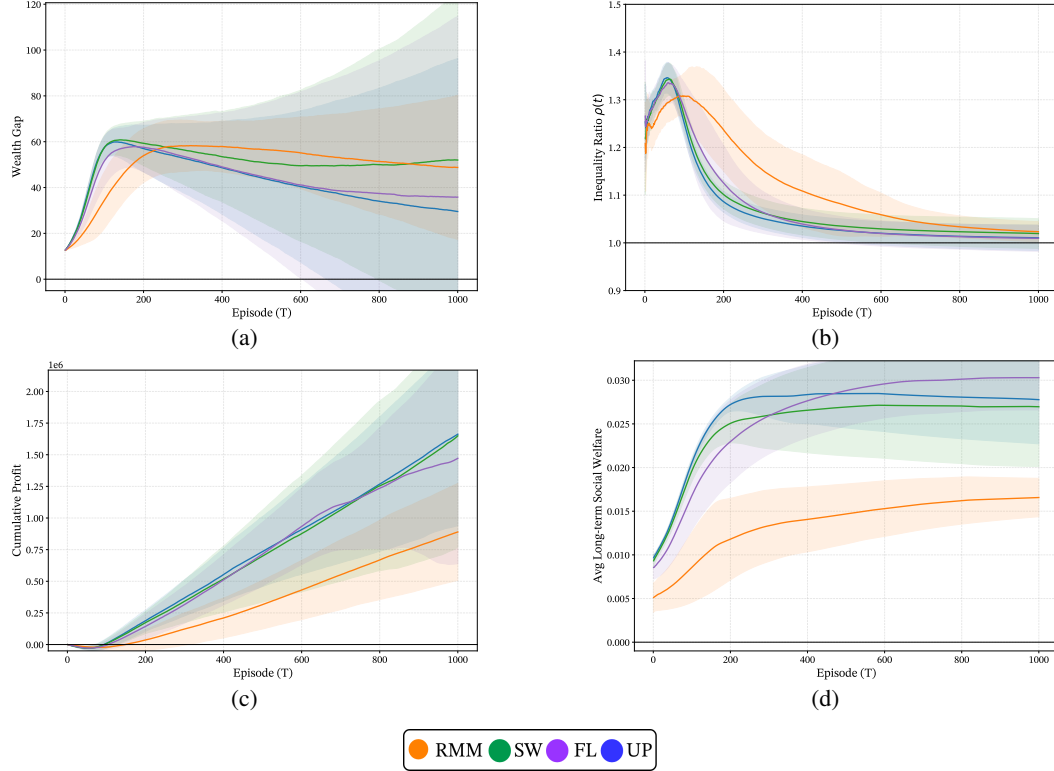


Figure 15: **RL: Two-Sided Fairness.** Long-term performance of RL across fairness objectives under Two-Sided fairness, tracked over $T = 1000$ episodes. Mean $\pm$ std across seeds.

larger than under DM fairness. The diamond shape reflects the alignment between the fairness constraint and the outcome metrics: targeting group wealth directly creates an incentive that partially propagates to all four dimensions simultaneously. The fairness objectives are more spread than under DM fairness, indicating that fairness objectives choice begins to matter once the fairness perspective operates at the wealth level rather than the approval rate level. The largest polygon is achieved by FL, which dominates on welfare and profit while maintaining moderate equality coverage, at the cost of the weakest wealth gap performance among the three fairness objectives shown.

Two-sided fairness. Figure 16c is the most distinctive of the three. The polygon is uniquely strong on the wealth gap axis, reaching the outer ring in a way that neither DM nor outcome fairness achieves. This is the only constraint that produces meaningful coverage on all four axes simultaneously, including the wealth gap, making it the closest to a balanced polygon across the full set of objectives. However this comes with two costs. First, welfare and profit are comparatively lower and more variable than under outcome fairness. Second, the spread between fairness objectives is the largest of any perspective: the polygons differ substantially in shape, with some fairness objectives trading equality for profit and others doing the reverse. Two-sided fairness therefore offers the broadest coverage of the utility space but also the least predictable performance profile, and the choice of fairness objective has the most consequential effect on which dimensions are prioritised.

Taken together, the three radar plots reveal a clear progression in the nature of the performance trade-off as the fairness perspective becomes more comprehensive. DM fairness produces a stable but asymmetric profile that ignores the wealth gap entirely. Outcome fairness produces a balanced profile that covers all four axes at moderate-to-strong levels. Two-sided fairness uniquely resolves the wealth gap but introduces instability in the other dimensions. No single constraint dominates on all four axes simultaneously, and the choice of fairness objective shapes not just the fairness outcomes but the entire performance geometry of the deployed policy.

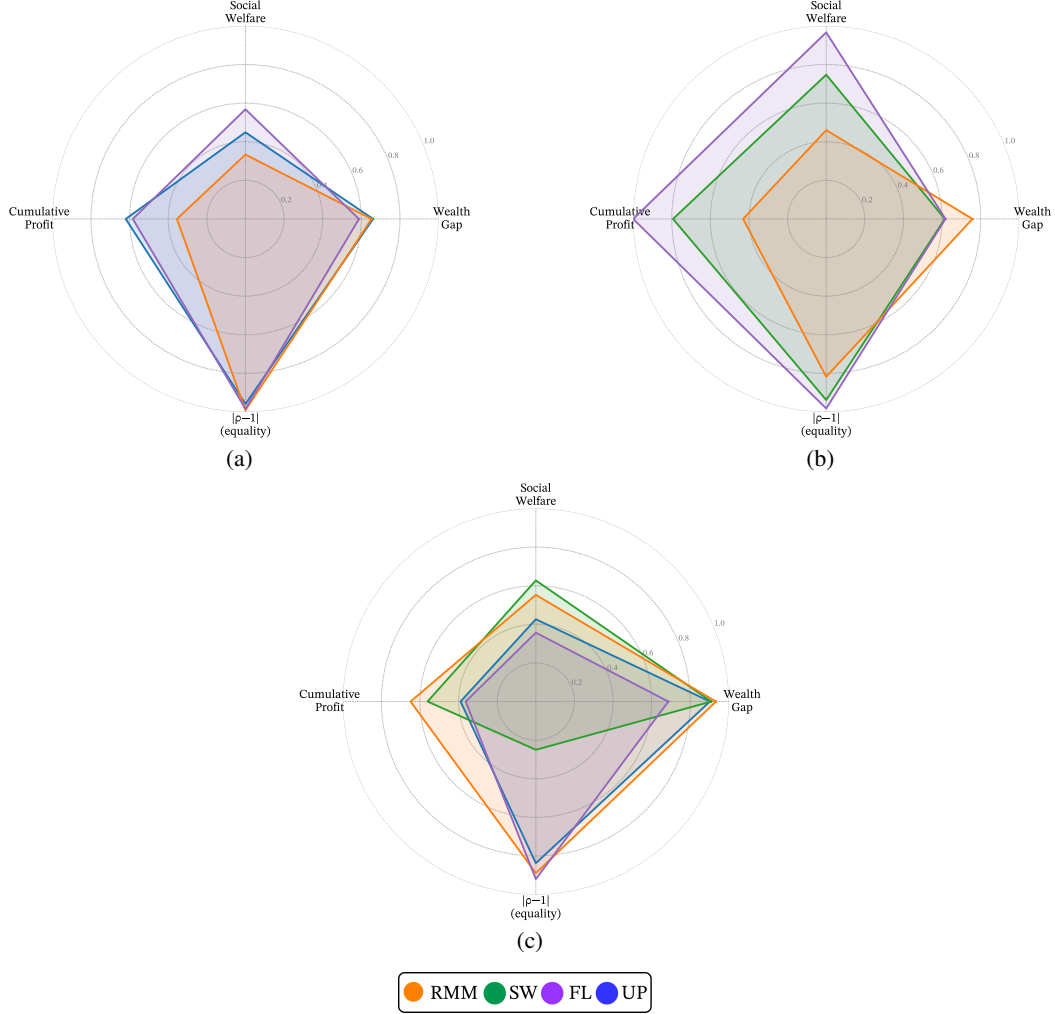


Figure 16: PERL: Radar Plots

E.3.4 MICRO-ANALYSIS: INCLUSIVITY AND DIVERSITY

Inter- and intra-group wealth diversity. Figure 4 and Figure 17 show Lorenz curves for cumulative loan fraction against cumulative population fraction, plotted at episodes 100, 400, 800, and 1000 (light to dark) for both demographic groups (red: male, blue: female) under DM and outcome fairness respectively. A curve closer to the diagonal indicates a more equal distribution of loans across the population within that group; a curve further from the diagonal indicates greater concentration among the highest-wealth individuals. The dashed line is the line of perfect equality.

A universal observation across all six plots is that curves move away from the diagonal as deployment progresses: regardless of fairness perspective or fairness objective, loan concentration within both groups increases over time. This is a direct manifestation of the Matthew effect. Early in deployment, wealth is relatively uniformly distributed and the policy approves a broad cross-section of applicants. As approved individuals accumulate wealth through loan repayments, they become increasingly likely to apply and be approved again, while lower-wealth individuals fall progressively further behind. The Lorenz curves record this compounding: each successive snapshot sits further from the diagonal, and no fairness perspective or fairness objective reverses this trajectory within the deployment horizon considered here.

DM fairness. The three fairness objectives produce qualitatively different distributional shapes. Under FL, the red and blue families are clearly separated throughout deployment, with the male group consistently receiving a more equal share of loans than the female group. The inter-group gap is large and persistent across all four episode snapshots, indicating that the Lagrangian penalty on

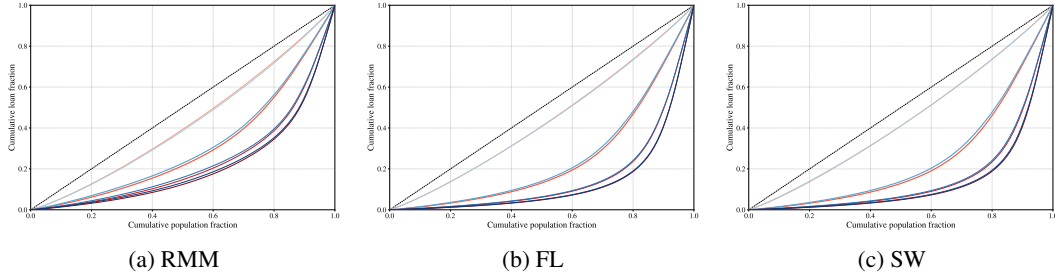


Figure 17: **Lorenz curves under outcome fairness (PERL)**. Red: male group, Blue: female group. Curves shown at episodes 100, 400, 800, and 1000 (light to dark). Dashed line is perfect equality.

profit disparity does not prevent the female group from being systematically more excluded from credit access as wealth stratifies. Under RMM, the red and blue families are much more closely interleaved, with smaller inter-group separation at every snapshot. However, this should not be interpreted as a more equitable distribution: as the histogram analysis confirms, RMM under DM fairness issues fewer total loans, and the relative skewness within each group persists at a similar level to other configurations. The apparent inter-group convergence reflects a shared reduction in loan volume rather than a genuine redistribution toward lower-wealth applicants. Under UP, the curves display visible kinks around the 0.4–0.6 population fraction, a feature not present in any other plot. These kinks indicate a threshold-driven lending pattern: approvals concentrate discontinuously at a particular wealth level, producing a step in the cumulative loan distribution rather than a smooth decay. This is consistent with the irregular reach rate trajectories observed for UP under DM fairness, and points to the unconstrained profit objective converging to a wealth-threshold policy that sharply distinguishes eligible from ineligible applicants.

Outcome fairness. The inter-group separation is generally larger under outcome fairness than under DM fairness for the same reward function. Under FL, the blue family is more tightly clustered across episode snapshots while the red family spreads more, suggesting that FL under outcome fairness stabilises the female group’s loan distribution while male loan concentration continues to grow. Under SW, the female group’s curves at late deployment are the closest to the diagonal of any configuration in either figure, indicating that SW under outcome fairness produces the most broadly distributed loans within the female group across the entire deployment horizon. This is consistent with SW’s objective of maximising aggregate group wealth, which under the performative feedback loop creates an incentive to maintain a wider base of loan recipients rather than concentrating entirely on the already-wealthy. Under RMM, the female group’s curves drift further from the diagonal relative to their starting position than under RMM+DM, suggesting that switching from DM to outcome fairness under RMM worsens within-group equity for the disadvantaged group despite reducing the between-group wealth gap at the aggregate level. This tension between aggregate fairness metrics and within-group distributional equity is not visible from the group-level statistics alone and is only revealed by the Lorenz curve analysis.

Inclusivity. Figure 5 shows the reach rate — the fraction of unique recipients per group per episode — for both demographic groups (red: male, blue: female) across all six configurations. Figure 18 and 19 shows the corresponding cumulative loan distributions, where each bar represents total loans received by an individual over the full deployment horizon, sorted by wealth index. Together these two views capture complementary aspects of inclusivity: the reach rate measures the breadth of credit access over time, while the histograms reveal the within-group concentration of that access.

A universal pattern holds across all reach rate plots: the reach rate rises in early deployment as the policy explores the population, peaks, and then declines as the Matthew effect takes hold and loans concentrate among repeat recipients. This decline is the inclusivity counterpart of the Lorenz curve drift observed above, and the histograms make the mechanism concrete: in every configuration, the cumulative loan distribution follows a strongly right-skewed shape where the top few individuals by wealth receive a disproportionate share of all loans issued over the deployment horizon. The solid fitted curve consistently exceeds the dashed reference in all twelve histograms, confirming that no configuration achieves a uniform distribution of credit access.

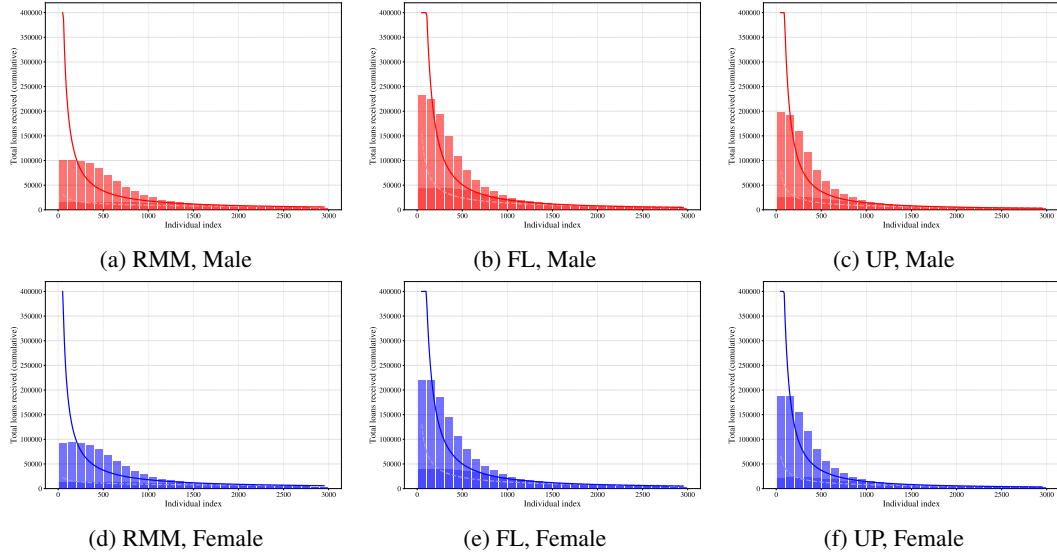


Figure 18: **Cumulative loan distributions under DM fairness (PERL)**. Each bar shows total loans received by an individual over the full deployment horizon, sorted by wealth index. Top row: male group. Bottom row: female group.

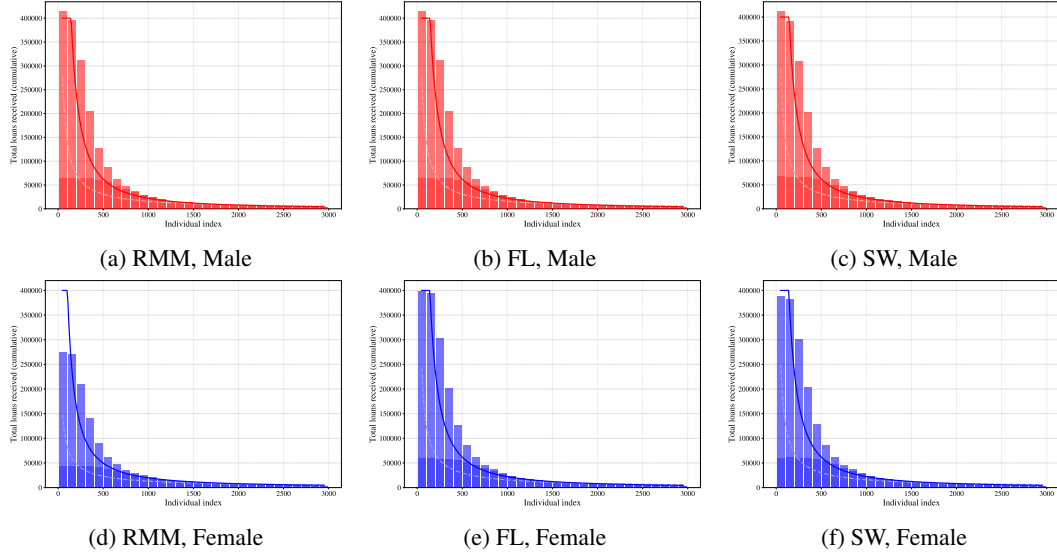


Figure 19: **Cumulative loan distributions under outcome fairness (PERL)**. Each bar shows total loans received by an individual over the full deployment horizon, sorted by wealth index. Top row: male group. Bottom row: female group.

Under DM fairness, the three fairness objectives differ in reach rate trajectory and in total loan volume, but not in the shape of within-group concentration. FL peaks highest among the DM configurations at approximately 0.35–0.36 around episode 150, with a sharp drop around episode 350–400 before stabilising near 0.14–0.15, suggesting a policy reorganisation point where the DM constraint becomes binding. The corresponding histograms show first-bar heights of approximately 220–230k for both groups, with a moderately steep decline. RMM peaks later and lower at approximately 0.27 around episode 200–250, with a smooth and gradual decline to approximately 0.17 by episode 1000. Its histograms show first-bar heights of only 95–100k, which is the lowest of any configuration, but the relative skewness of the distribution persists: fewer loans are issued in total, but the top individuals still receive a disproportionate share within each group. This reflects RMM’s

conservative aggregate approval behaviour under DM fairness rather than any genuine redistribution toward lower-wealth applicants. UP peaks lowest at approximately 0.20 with the widest seed-level variance, including negative values for the female group in early deployment, and produces intermediate histogram heights of approximately 190–200k.

Under outcome fairness, all three configurations achieve higher peak reach rates than their DM counterparts, but this comes at the cost of sharper within-group concentration. FL and SW both reach peak rates of approximately 0.43–0.45 around episode 120–130 with near-perfect group parity and very tight credible intervals, and their histograms are nearly identical: first bars hit or exceed the 400k axis limit, followed by a very steep cliff, making these the most concentrated distributions across all twelve plots. The close correspondence between FL and SW here is consistent with their shared objective of maximising total group wealth, which under performative feedback rewards repeated lending to the already-wealthy. RMM under outcome fairness occupies an intermediate position: it peaks near 0.35 in reach rate but then exhibits a step-drop around episode 250–300 and ends at the lowest terminal reach rate of all configurations at approximately 0.10–0.11. Its histograms show first-bar heights of approximately 270–290k, intermediate between RMM+DM and the FL/SW outcome configurations, with male and female groups nearly identical in shape throughout.

F COMPUTATIONAL RESOURCES

All experiments reported in this paper were run on a single workstation with two Intel Xeon Gold 6134 CPUs (8 cores / 16 threads per socket, base clock 3.20 GHz; 32 logical cores total) and 128 GB of RAM, running Linux. No GPU was used: the policy networks are small and the per-step bottleneck is the environment simulation rather than the gradient computation, so a CPU-only setup keeps cost and parallelism favourable. Parallelism across runs is exposed through a Python `multiprocessing` pool. As an indicative wall-clock, a full sweep across agent types, fairness objectives, fairness perspectives, and 10 seeds, at $T = 1000$ deploy episodes each, completes in approximately 20 hours on this machine.